

University of Minnesota
School of Physics and Astronomy

2026 Spring Physics 8902
Elementary Particle Physics II
Assignment Solution

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Problem Set 1 Due 11am, Monday, February 2

Question 1

Show that the charge-lowering weak current of the form

$$J^\mu = \bar{e}\gamma^\mu \frac{1}{2}(1 - \gamma^5)\nu_e \quad (1)$$

involves only left-handed electrons (or right-handed positrons). In the relativistic limit ($v \approx c$) show that the electrons have negative helicity.

Answer

We start with the Dirac spinor:

$$\psi = \begin{pmatrix} \psi_L \\ \psi_R \end{pmatrix}, \quad (2)$$

where ψ_L and ψ_R are the left-handed and right-handed components, respectively. To be more specific, here the left-handed and right-handed are referring to chirality. Besides, this notation is called Weyl or chiral representation. In this representation, the gamma matrices are expressed as:

$$\gamma^0 = \begin{pmatrix} 0 & I \\ I & 0 \end{pmatrix}, \quad \gamma^i = \begin{pmatrix} 0 & \sigma^i \\ -\sigma^i & 0 \end{pmatrix}, \quad \gamma^5 = \begin{pmatrix} -I & 0 \\ 0 & I \end{pmatrix}, \quad (3)$$

where I is the 2×2 identity matrix and σ^i are the Pauli matrices. The projection operators for left-handed and right-handed components are given by:

$$P_L = \frac{1}{2}(1 - \gamma^5), \quad P_R = \frac{1}{2}(1 + \gamma^5). \quad (4)$$

Then the projection operator P_L can be explicitly written as:

$$P_L = \frac{1}{2} \begin{pmatrix} 2I & 0 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} I & 0 \\ 0 & 0 \end{pmatrix}. \quad (5)$$

Applying this operator to the Dirac spinor ψ , we have:

$$P_L \psi = \begin{pmatrix} I & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} \psi_L \\ \psi_R \end{pmatrix} = \begin{pmatrix} \psi_L \\ 0 \end{pmatrix}. \quad (6)$$

Besides, we have this relation:

$$\bar{\psi}\gamma^\mu P_L\psi = \bar{\psi}P_R\gamma^\mu\psi, \quad (7)$$

where we used the fact that $\gamma^\mu P_L = P_R\gamma^\mu$. The $\bar{\psi}$ is defined as $\bar{\psi} = \psi^\dagger\gamma^0$. Therefore, we have:

$$J^\mu = \bar{e}\gamma^\mu P_L\nu_e = \bar{e}\gamma^\mu P_L P_L\nu_e \quad (8)$$

$$= \bar{e}P_R\gamma^\mu P_L\nu_e = e^\dagger\gamma^0 P_R\gamma^\mu P_L\nu_e \quad (9)$$

$$= e^\dagger P_L\gamma^0\gamma^\mu P_L\nu_e \quad (10)$$

$$= (P_L e)^\dagger\gamma^0\gamma^\mu(P_L\nu_e) \quad (11)$$

$$= \bar{e}_L\gamma^\mu\nu_{eL}. \quad (12)$$

This shows that the weak current J^μ only involves left-handed electrons (or right-handed positrons). Besides, it also involves left-handed neutrinos.

Next, we start from the Dirac equation, the solution for Dirac spinor $u(p)$ with momentum p and mass m can be expressed as:

$$u(p) = \begin{pmatrix} \sqrt{p \cdot \sigma} \xi_s \\ \sqrt{p \cdot \bar{\sigma}} \xi_s \end{pmatrix}, \quad (13)$$

where $\sigma^\mu = (I, \sigma^i)$, $\bar{\sigma}^\mu = (I, -\sigma^i)$, and ξ_s is a two-component spinor representing the spin state:

$$\xi_+ = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \xi_- = \begin{pmatrix} 0 \\ 1 \end{pmatrix}. \quad (14)$$

For simplicity, we consider the electron moving along the z -axis, so the four-momentum can be written as:

$$p^\mu = (E, 0, 0, p_z), \quad (15)$$

where $E = \sqrt{p_z^2 + m^2}$. Hence, we have:

$$u_+(p) = \begin{pmatrix} \sqrt{E - p_z} \\ 0 \\ \sqrt{E + p_z} \\ 0 \end{pmatrix}, \quad u_-(p) = \begin{pmatrix} 0 \\ \sqrt{E + p_z} \\ 0 \\ \sqrt{E - p_z} \end{pmatrix}. \quad (16)$$

In particular, the left-handed component of the Dirac spinor can be obtained by applying the projection

operator P_L :

$$P_L u_+(p) = \begin{pmatrix} \sqrt{E - p_z} \\ 0 \\ 0 \\ 0 \end{pmatrix}, \quad P_L u_-(p) = \begin{pmatrix} 0 \\ \sqrt{E + p_z} \\ 0 \\ 0 \end{pmatrix}. \quad (17)$$

Now the helicity operator is defined as:

$$h = \frac{\vec{\sigma} \cdot \vec{p}}{|\vec{p}|} = \Sigma^3 = \begin{pmatrix} \sigma^3 & 0 \\ 0 & \sigma^3 \end{pmatrix} \frac{p_z}{|p_z|} \quad (18)$$

Now we can apply this operator to act on the solution with left-handed spinor, we have:

$$h P_L u_-(p) = \begin{pmatrix} \sigma^3 & 0 \\ 0 & \sigma^3 \end{pmatrix} \frac{p_z}{|p_z|} \begin{pmatrix} \sqrt{E - p_z} \\ 0 \\ 0 \\ 0 \end{pmatrix} = \frac{p_z}{|p_z|} \begin{pmatrix} \sqrt{E - p_z} \\ 0 \\ 0 \\ 0 \end{pmatrix}, \quad (19)$$

and

$$h P_L u_-(p) = \begin{pmatrix} \sigma^3 & 0 \\ 0 & \sigma^3 \end{pmatrix} \frac{p_z}{|p_z|} \begin{pmatrix} 0 \\ \sqrt{E + p_z} \\ 0 \\ 0 \end{pmatrix} = -\frac{p_z}{|p_z|} \begin{pmatrix} 0 \\ \sqrt{E + p_z} \\ 0 \\ 0 \end{pmatrix}. \quad (20)$$

In the relativistic limit, we have $p_z = \pm E$. For $p_z = +E$, only $u_-(p)$ survives after the projection, and we have:

$$h P_L u_-(p) = -P_L u_-(p), \quad (21)$$

which means the helicity is negative. For $p_z = -E$, only $u_+(p)$ survives after the projection, and we have:

$$h P_L u_+(p) = -P_L u_+(p), \quad (22)$$

which also means the helicity is negative. **Therefore, in the relativistic limit, the left-handed electrons have negative helicity.** $\square$

Question 2

Weak decays of leptons

- (a) Calculate the muon total decay width, $\Gamma(\mu^- \rightarrow e^- \bar{\nu}_e \nu_\mu)$, accounting for the finite mass of the electron.
- (b) Use this result to determine the numerical value of the ratio

$$R = \frac{\Gamma(\tau^- \rightarrow \mu^- \bar{\nu}_\mu \nu_\tau)}{\Gamma(\tau^- \rightarrow e^- \bar{\nu}_e \nu_\tau)}. \quad (23)$$

Compare the theoretical prediction for this ratio with the experimental value.

Answer

(a) We can start from the differential decay width for the muon decay process $\mu^- \rightarrow e^- \bar{\nu}_e \nu_\mu$, which is given by (this is allowed in the lecture notes):

$$d\Gamma = \frac{1}{12\pi} G_F^2 \left[3(m_\mu^2 + m_e^2)E - 4m_\mu^2 E - 2m_\mu m_e^2 \right] |\vec{p}| dE, \quad (24)$$

where G_F is the Fermi coupling constant, m_μ and m_e are the masses of muon and electron, respectively. Here E and $\vec{p}$ are the energy and momentum of the electron in the final state. Hence, we can integrate over the electron energy E to get the total decay width. The range of E is from m_e to $\frac{m_\mu^2 + m_e^2}{2m_\mu}$. The upper limit can be derived from the energy-momentum conservation in the muon rest frame:

$$m_\mu = E + E_{\bar{\nu}_e} + E_{\nu_\mu} \geq E + |\vec{p}_{\bar{\nu}_e}| + |\vec{p}_{\nu_\mu}| \geq E + |\vec{p}_{\bar{\nu}_e} + \vec{p}_{\nu_\mu}| = E + |\vec{p}| = E + \sqrt{E^2 - m_e^2} \quad (25)$$

$$\implies E \leq \frac{m_\mu^2 + m_e^2}{2m_\mu}. \quad (26)$$

Therefore, we have:

$$\Gamma = \frac{1}{12\pi} G_F^2 \int_{m_e}^{\frac{m_\mu^2 + m_e^2}{2m_\mu}} \left[3(m_\mu^2 + m_e^2)E - 4m_\mu^2 E - 2m_\mu m_e^2 \right] \sqrt{E^2 - m_e^2} dE \quad (27)$$

$$= \frac{1}{12\pi} G_F^2 \frac{8m_e^6 m_\mu^2 - 8m_e^2 m_\mu^6 + 24m_e^4 m_\mu^4 \log\left(\frac{m_\mu}{m_e}\right) - m_e^8 + m_\mu^8}{16m_\mu^3}, \quad \text{by using Mathematica.} \quad (28)$$

$$\approx \frac{G_F^2 m_\mu^5}{192\pi^3} \left[1 - 8 \frac{m_e^2}{m_\mu^2} + \mathcal{O}\left(\frac{m_e^4}{m_\mu^4}\right) \right]. \quad (29)$$

This is the total decay width of muon accounting for the finite mass of the electron.

(b) Now we can use the result from part (a) to calculate the ratio:

$$R_{\text{th}} = \frac{\Gamma(\tau^- \rightarrow \mu^- \bar{\nu}_\mu \nu_\tau)}{\Gamma(\tau^- \rightarrow e^- \bar{\nu}_e \nu_\tau)} \quad (30)$$

$$= \frac{1 - 8 \frac{m_\mu^2}{m_\tau^2} + \mathcal{O}\left(\frac{m_\mu^4}{m_\tau^4}\right)}{1 - 8 \frac{m_e^2}{m_\tau^2} + \mathcal{O}\left(\frac{m_e^4}{m_\tau^4}\right)} \quad (31)$$

$$\approx (1 - 8 \frac{m_\mu^2}{m_\tau^2})(1 + 8 \frac{m_e^2}{m_\tau^2}) \quad (32)$$

$$\approx 1 - 8 \frac{m_\mu^2 - m_e^2}{m_\tau^2}. \quad (33)$$

Using the known values of the masses: $m_e = 0.511$ MeV, $m_\mu = 105.66$ MeV, and $m_\tau = 1776.86$ MeV, we can calculate the numerical value of $R_{\text{th}} = 0.971712$. The experimental value is given by

$$R_{\text{exp}} = \frac{B(\tau^- \rightarrow \mu^- \bar{\nu}_\mu \nu_\tau)}{B(\tau^- \rightarrow e^- \bar{\nu}_e \nu_\tau)} = \frac{0.1739}{0.1782} = 0.97587. \quad (34)$$

The theoretical prediction is in good agreement with the experimental value, with a small difference that can be attributed to higher-order corrections and experimental uncertainties. $\square$

Problem Set 2 Due 11am, Monday, February 16

Question 1

Weak decay of pions

- (a) Find the electron energy spectrum $d\Gamma/dE_e$ for the decay $\pi^- \rightarrow \pi^0 + e^- + \bar{\nu}_e$ in the π^- rest frame keeping $m_e \neq 0$ (take $m_\nu = 0$). Assume the hadronic current is dominated by $f_+(0)$ and neglect radiative corrections. Perform the phase-space integration by integrating over the π^0 and $\bar{\nu}_e$ momenta (i.e. treat E_e as the only observed variable). Give the kinematic endpoints and verify the $m_e \rightarrow 0$ limit.
- (b) Using the electron energy spectrum obtained in part (a), integrate over E_e to extract the leading correction of order m_e^2/Δ^2 to the total decay rate. Write the result in the form

$$\Gamma(\pi^- \rightarrow \pi^0 + e^- + \bar{\nu}_e) = |V_{ud}|^2 \frac{G_F^2 \Delta^5}{30\pi^3} \left(1 - a \frac{\Delta}{m_\pi} - b \frac{m_e^2}{\Delta^2} \right), \quad (35)$$

where $\Delta = m_{\pi^-} - m_{\pi^0}$, and neglecting higher-order terms in Δ/m_π and m_e^2/Δ^2 . In the lectures it was shown that $a = 3/2$. Determine the coefficient b .

Answer

(a)

Let us denote the momenta of π^- , π^0 , e^- , and $\bar{\nu}_e$ as p , p' , k , and k' , respectively. The decay amplitude can be written as

$$\mathcal{M} = \langle \pi^0(p') e^-(k) \bar{\nu}_e(k') | \mathcal{H}_W | \pi^-(p) \rangle = -\frac{G_F}{\sqrt{2}} V_{ud} \langle \pi^0(p') | \bar{d} \gamma^\mu u | \pi^-(p) \rangle \langle e^-(k) \bar{\nu}_e(k') | \bar{e} \gamma_\mu (1 - \gamma_5) \nu_e | 0 \rangle. \quad (36)$$

The hadronic matrix element can be parameterized as

$$\langle \pi^0(p') | \bar{d} \gamma^\mu u | \pi^-(p) \rangle = f_+(q^2)(p + p')^\mu + f_-(q^2)(p - p')^\mu \approx f_+(0)(p + p')^\mu = \sqrt{2}(p + p')^\mu, \quad (37)$$

where $q = p - p'$. Neglecting radiative corrections and using the fact that $f_+(0)$ dominates, we can approximate $f_+(q^2) \approx f_+(0) = \sqrt{2}$. The leptonic matrix element can be evaluated using standard techniques, yielding

$$\langle e^-(k) \bar{\nu}_e(k') | \bar{e} \gamma_\mu (1 - \gamma_5) \nu_e | 0 \rangle = \bar{u}(k) \gamma_\mu (1 - \gamma_5) v(k'). \quad (38)$$

Now the amplitude can be expressed as

$$\mathcal{M} = -\frac{G_F}{\sqrt{2}} V_{ud} \sqrt{2} (p + p')^\mu \bar{u}(k) \gamma_\mu (1 - \gamma_5) v(k') \quad (39)$$

$$= -G_F V_{ud} (p + p')^\mu \bar{u}(k) \gamma_\mu (1 - \gamma_5) v(k'). \quad (40)$$

Then the squared amplitude, summed over final spins, is given by

$$\langle |\mathcal{M}|^2 \rangle = \sum_{\text{spins}} |\mathcal{M}|^2 = G_F^2 |V_{ud}|^2 (p + p')^\mu (p + p')^\nu \bar{u}_{s_1}(k) \gamma_\mu (1 - \gamma_5) v_{s_2}(k') \bar{v}_{s_2}(k') \gamma_\nu (1 - \gamma_5) u_{s_1}(k) \quad (41)$$

$$= G_F^2 |V_{ud}|^2 (p + p')^\mu (p + p')^\nu \text{Tr} \left[(\not{k} + m_e) \gamma_\mu (1 - \gamma_5) \not{k}' \gamma_\nu (1 - \gamma_5) \right] \quad (42)$$

$$= G_F^2 |V_{ud}|^2 (p + p')^\mu (p + p')^\nu k^\alpha k'^\beta \text{Tr} \left[(\gamma_\alpha + 1 m_e) \gamma_\mu (1 - \gamma_5) \gamma_\beta \gamma_\nu (1 - \gamma_5) \right]. \quad (43)$$

where we have used the spin sum identities for the electron and neutrino:

$$\sum_{s_1} u_{s_1}(k) \bar{u}_{s_1}(k) = \not{k} + m_e, \quad \sum_{s_2} v_{s_2}(k') \bar{v}_{s_2}(k') = \not{k}' - m_\nu \approx \not{k}'. \quad (44)$$

We provide the full set of trace identities:

$$\text{Tr}(\gamma^\mu \gamma^\nu) = 4g^{\mu\nu}, \quad \text{Tr}(\gamma^\mu \gamma^\nu \gamma_5) = 0, \quad \text{Tr}(\gamma^\mu \gamma^\nu \gamma^\rho \gamma^\sigma) = 4(g^{\mu\nu} g^{\rho\sigma} - g^{\mu\rho} g^{\nu\sigma} + g^{\mu\sigma} g^{\nu\rho}), \quad (45)$$

$$\text{Tr}(\gamma^\mu \gamma^\nu \gamma^\rho \gamma^\sigma \gamma_5) = 4i\epsilon^{\mu\nu\rho\sigma}, \quad \text{Tr}(\text{odd number of } \gamma_5) = 0. \quad (46)$$

Hence,

$$\text{Tr} \left[(\gamma_\alpha + 1 m_e) \gamma_\mu (1 - \gamma_5) \gamma_\beta \gamma_\nu (1 - \gamma_5) \right] \quad (47)$$

$$= \text{Tr} \left[\gamma_\alpha \gamma_\mu (1 - \gamma_5) \gamma_\beta \gamma_\nu (1 - \gamma_5) \right] + m_e \text{Tr} \left[\gamma_\mu (1 - \gamma_5) \gamma_\beta \gamma_\nu (1 - \gamma_5) \right] \quad (48)$$

$$= \text{Tr} \left[\gamma_\alpha \gamma_\mu \gamma_\beta \gamma_\nu \right] + \text{Tr} \left[\gamma_\alpha \gamma_\mu \gamma_5 \gamma_\beta \gamma_\nu \gamma_5 \right] - \text{Tr} \left[\gamma_\alpha \gamma_\mu \gamma_5 \gamma_\beta \gamma_\nu \right] - \text{Tr} \left[\gamma_\alpha \gamma_\mu \gamma_\beta \gamma_\nu \gamma_5 \right] \quad (49)$$

$$+ m_e \text{Tr} \left[\gamma_\mu \gamma_\beta \gamma_\nu \right] - m_e \text{Tr} \left[\gamma_\mu \gamma_5 \gamma_\beta \gamma_\nu \right] - m_e \text{Tr} \left[\gamma_\mu \gamma_\beta \gamma_\nu \gamma_5 \right] + m_e \text{Tr} \left[\gamma_\mu \gamma_5 \gamma_\beta \gamma_\nu \gamma_5 \right] \quad (50)$$

$$= 2 \text{Tr} \left[\gamma_\alpha \gamma_\mu \gamma_\beta \gamma_\nu \right] - 2 \text{Tr} \left[\gamma_\alpha \gamma_\mu \gamma_\beta \gamma_\nu \gamma_5 \right] \quad (51)$$

$$= 8(g_{\alpha\mu} g_{\beta\nu} - g_{\alpha\beta} g_{\mu\nu} + g_{\alpha\nu} g_{\mu\beta}) - 8i\epsilon_{\alpha\mu\beta\nu}. \quad (52)$$

Actually, the term $\epsilon_{\alpha\mu\beta\nu}$ will vanish later when contracted with $(p + p')^\mu (p + p')^\nu k^\alpha k'^\beta$, since $p + p'$ is symmetric in μ and ν , while $\epsilon_{\alpha\mu\beta\nu}$ is antisymmetric in μ and ν . Therefore, we can ignore the second

term and write

$$\langle |\mathcal{M}|^2 \rangle = G_F^2 |V_{ud}|^2 (p + p')^\mu (p + p')^\nu k^\alpha k'^\beta 8(g_{\alpha\mu}g_{\beta\nu} - g_{\alpha\beta}g_{\mu\nu} + g_{\alpha\nu}g_{\mu\beta}) \quad (53)$$

$$= 8G_F^2 |V_{ud}|^2 (p + p')^\mu (p + p')^\nu \left(k_\mu k'_\nu - (k \cdot k') g_{\mu\nu} + k_\nu k'_\mu \right) \quad (54)$$

$$= 8G_F^2 |V_{ud}|^2 \left[2(p + p') \cdot k (p + p') \cdot k' - (p + p')^2 (k \cdot k') \right] \quad (55)$$

We can assume the 4-momenta in the π^- rest frame as

$$p = (m_{\pi^-}, \mathbf{0}), \quad p' = (E_{\pi^0}, -(\mathbf{k} + \mathbf{k}') = \mathbf{p}'), \quad k = (E_e, \mathbf{k}), \quad k' = (E_\nu, \mathbf{k}'), \quad (56)$$

where $E_{\pi^0} = \sqrt{|\mathbf{p}'|^2 + m_{\pi^0}^2}$, $E_e = \sqrt{|\mathbf{k}|^2 + m_e^2}$, and $E_\nu = |\mathbf{k}'|$. We can define $q = p - p' = k + k'$ and have following relations:

$$(p + p') \cdot k = (2p - (k + k')) \cdot k = 2p \cdot k - k^2 - (k \cdot k') = 2m_{\pi^-} E_e - m_e^2 - (k \cdot k'), \quad (57)$$

$$(p + p') \cdot k' = (2p - (k + k')) \cdot k' = 2p \cdot k' - k'^2 - (k \cdot k') = 2m_{\pi^-} E_\nu - (k \cdot k'), \quad (58)$$

$$(p + p')^2 = (2p - (k + k'))^2 = 4m_{\pi^-}^2 - 4m_{\pi^-}(E_e + E_\nu) + (k + k')^2 \quad (59)$$

$$= 4m_{\pi^-}^2 - 4m_{\pi^-}(E_e + E_\nu) + m_e^2 + 2(k \cdot k'). \quad (60)$$

Since the m_{π^-} is much larger than the energy of the final state particles (except for the π^0), we can further approximate $E_e + E_\nu \approx m_{\pi^-} - m_{\pi^0} = \Delta$. Hence, we have

$$2[(p + p') \cdot k][(p + p') \cdot k'] = 2(2m_{\pi^-} E_e - m_e^2 - (k \cdot k'))(2m_{\pi^-} E_\nu - (k \cdot k')) \quad (61)$$

$$= 8m_{\pi^-}^2 E_e E_\nu - 4m_{\pi^-}(E_e + E_\nu)(k \cdot k') - 4m_e^2 m_{\pi^-} E_\nu + 2m_e^2 (k \cdot k') + 2(k \cdot k')^2 \quad (62)$$

$$= 8m_{\pi^-}^2 E_e E_\nu - 4m_{\pi^-} \Delta (k \cdot k') - 4m_e^2 m_{\pi^-} E_\nu + 2m_e^2 (k \cdot k') + 2(k \cdot k')^2, \quad (63)$$

$$(p + p')^2 (k \cdot k') = (4m_{\pi^-}^2 - 4m_{\pi^-} \Delta + m_e^2 + 2(k \cdot k'))(k \cdot k') \quad (64)$$

$$= 4m_{\pi^-}^2 (k \cdot k') - 4m_{\pi^-} \Delta (k \cdot k') + m_e^2 (k \cdot k') + 2(k \cdot k')^2. \quad (65)$$

Putting everything together, we can express the squared amplitude as

$$\langle |\mathcal{M}|^2 \rangle = 8G_F^2 |V_{ud}|^2 \left[8m_{\pi^-}^2 E_e E_\nu - 4m_{\pi^-} \Delta (k \cdot k') - 4m_e^2 m_{\pi^-} E_\nu + 2m_e^2 (k \cdot k') + 2(k \cdot k')^2 \right] \quad (66)$$

$$- 4m_{\pi^-}^2 (k \cdot k') + 4m_{\pi^-} \Delta (k \cdot k') - m_e^2 (k \cdot k') - 2(k \cdot k')^2 \quad (67)$$

$$= 8G_F^2 |V_{ud}|^2 \left[8m_{\pi^-}^2 E_e E_\nu - 4m_{\pi^-}^2 (k \cdot k') - 4m_e^2 m_{\pi^-} E_\nu + m_e^2 (k \cdot k') \right]. \quad (68)$$

Now, we can express $k \cdot k'$ in terms of the energies and the angle between the electron and neutrino

momenta. Let θ be the angle between $\mathbf{k}$ and $\mathbf{k}'$, then we have:

$$k \cdot k' = E_e E_\nu - \mathbf{k} \cdot \mathbf{k}' = E_e E_\nu - |\mathbf{k}| |\mathbf{k}'| \cos \theta = E_e E_\nu - |\mathbf{k}| |\mathbf{k}'| \cos \theta \quad (69)$$

$$= E_e E_\nu (1 - \beta_e \cos \theta), \quad \text{where } \beta_e = \frac{|\mathbf{k}|}{E_e} = \sqrt{1 - \frac{m_e^2}{E_e^2}}. \quad (70)$$

The squared amplitude can be further simplified as

$$\langle |\mathcal{M}|^2 \rangle = 8G_F^2 |V_{ud}|^2 \left[8m_{\pi^-}^2 E_e E_\nu - 4m_{\pi^-}^2 E_e E_\nu (1 - \beta_e \cos \theta) - 4m_e^2 m_{\pi^-} E_\nu + m_e^2 E_e E_\nu (1 - \beta_e \cos \theta) \right] \quad (71)$$

$$= 8G_F^2 |V_{ud}|^2 \left[4m_{\pi^-}^2 E_e E_\nu (1 + \beta_e \cos \theta) - 4m_e^2 m_{\pi^-} E_\nu + m_e^2 E_e E_\nu (1 - \beta_e \cos \theta) \right] \quad (72)$$

$$= 32G_F^2 |V_{ud}|^2 m_{\pi^-}^2 \left[E_e E_\nu (1 + \beta_e \cos \theta) - \frac{m_e^2}{m_{\pi^-}} E_\nu + \frac{m_e^2}{4m_{\pi^-}^2} E_e E_\nu (1 - \beta_e \cos \theta) \right]. \quad (73)$$

Besides,

$$q^2 = (k + k')^2 = m_e^2 + 2k \cdot k' = m_e^2 + 2E_e E_\nu (1 - \beta_e \cos \theta). \quad (74)$$

$$\implies E_e E_\nu (1 - \beta_e \cos \theta) = \frac{q^2 - m_e^2}{2}. \quad (75)$$

$$\implies E_e E_\nu (1 + \beta_e \cos \theta) = 2E_e E_\nu - E_e E_\nu (1 - \beta_e \cos \theta) = 2E_e E_\nu - \frac{q^2 - m_e^2}{2}. \quad (76)$$

Therefore, the squared amplitude can be expressed as

$$\langle |\mathcal{M}|^2 \rangle = 32G_F^2 |V_{ud}|^2 m_{\pi^-}^2 \left[2E_e E_\nu - \frac{q^2 - m_e^2}{2} - \frac{m_e^2}{m_{\pi^-}} E_\nu + \frac{m_e^2}{4m_{\pi^-}^2} \frac{q^2 - m_e^2}{2} \right] \quad (77)$$

The decay rate can be calculated using the standard formula for three-body decays:

$$d\Gamma = \frac{1}{2m_{\pi^-}} \langle |\mathcal{M}|^2 \rangle d\tau_3, \quad (78)$$

where $d\tau_3$ is the three-body phase space element, given by splitting formulas:

$$d\tau_3 = d\tau_2(p \rightarrow p' + q) \frac{dq^2}{2\pi} d\tau_2(q \rightarrow k + k'), \quad (79)$$

where $q = k + k'$ is the total momentum of the lepton pair. The two-body phase space elements can be

expressed as

$$d\tau_2(p \rightarrow p' + q) = (2\pi)^4 \delta^4(p - p' - q) \frac{d^3 p'}{(2\pi)^3 2E_{p'}} \frac{d^3 q}{(2\pi)^3 2E_q}, \quad (80)$$

$$d\tau_2(q \rightarrow k + k') = (2\pi)^4 \delta^4(q - k - k') \frac{d^3 k}{(2\pi)^3 2E_k} \frac{d^3 k'}{(2\pi)^3 2E_{k'}}. \quad (81)$$

Now, let's try to integrate over the π^0 and $\bar{\nu}_e$ momenta. Since there is no p' in the squared amplitude, we can first integrate over p' using the delta function in $d\tau_2(p \rightarrow p' + q)$, which gives

$$\int d\tau_2(p \rightarrow p' + q) = \int (2\pi)^4 \delta^4(p - p' - q) \frac{d^3 p'}{(2\pi)^3 2E_{p'}} \frac{d^3 q}{(2\pi)^3 2E_q} \quad (82)$$

$$= \int (2\pi)^4 \delta(m_{\pi^-} - E_{p'} - E_q) \frac{1}{(2\pi)^3 2E_{p'}} \frac{d^3 q}{(2\pi)^3 2E_q} \quad (83)$$

$$= \int (2\pi)^4 \delta(m_{\pi^-} - E_{p'} - E_q) \frac{1}{(2\pi)^3 2E_{p'}} \frac{4\pi |\mathbf{q}|^2 d|\mathbf{q}|}{(2\pi)^3 2E_q} \quad (84)$$

$$= \frac{1}{4\pi} \int \delta(m_{\pi^-} - E_{p'} - E_q) \frac{|\mathbf{q}|^2 d|\mathbf{q}|}{E_{p'} E_q} \quad (85)$$

$$= \frac{1}{4\pi} \frac{|\mathbf{q}|}{m_{\pi^-}} \quad \text{with } |\mathbf{q}| = \frac{\sqrt{(m_{\pi^-}^2 - (m_{\pi^0} + \sqrt{q^2})^2)(m_{\pi^-}^2 - (m_{\pi^0} - \sqrt{q^2})^2)}}{2m_{\pi^-}}. \quad (86)$$

We can further simplify the expression for $|\mathbf{q}|$ by noting that $\Delta = m_{\pi^-} - m_{\pi^0}$ is small compared to m_{π^-} , and $m_{\pi^0} \approx m_{\pi^-}$, hence we can approximate $|\mathbf{q}|$ as

$$|\mathbf{q}| = \frac{\underbrace{\sqrt{m_{\pi^-} + m_{\pi^0} - \sqrt{q^2}}}_{\approx \sqrt{2m_{\pi^-}}} \underbrace{\sqrt{m_{\pi^-} - m_{\pi^0} - \sqrt{q^2}}}_{=\sqrt{\Delta - \sqrt{q^2}}} \underbrace{\sqrt{m_{\pi^-} + m_{\pi^0} + \sqrt{q^2}}}_{\approx \sqrt{2m_{\pi^-}}} \underbrace{\sqrt{m_{\pi^-} - m_{\pi^0} + \sqrt{q^2}}}_{=\sqrt{\Delta + \sqrt{q^2}}}}{2m_{\pi^-}} \approx \sqrt{\Delta^2 - q^2}. \quad (87)$$

Now, we can express the three-body phase space element as

$$d\tau_3 = d\tau_2(p \rightarrow p' + q) \frac{dq^2}{2\pi} d\tau_2(q \rightarrow k + k') \quad (88)$$

$$= \frac{1}{4\pi} \frac{\sqrt{\Delta^2 - q^2}}{m_{\pi^-}} \frac{dq^2}{2\pi} d\tau_2(q \rightarrow k + k'). \quad (89)$$

Next, we can integrate over the neutrino momentum k' using the delta function in $d\tau_2(q \rightarrow k + k')$, and we consider the q rest frame to perform the integration over the electron momentum k . In the q rest frame, we have (denoted by a star):

$$q = (q^0, \mathbf{0}), \quad k = (E_e^*, \mathbf{k}^*), \quad k' = (E_\nu^*, -\mathbf{k}^*). \quad (90)$$

The energies of the electron and neutrino in the q rest frame can be expressed as

$$\int d\tau_2(q \rightarrow k + k') = \int (2\pi)^4 \delta^4(q - k - k') \frac{d^3k}{(2\pi)^3 2E_k} \frac{d^3k'}{(2\pi)^3 2E_{k'}} \quad (91)$$

$$= \int (2\pi)^4 \delta(q^0 - E_e^* - E_\nu^*) \frac{d^3k^*}{(2\pi)^3 2E_e^*} \frac{1}{(2\pi)^3 2E_\nu^*} \quad (92)$$

$$= \frac{1}{16\pi^2} \int \delta(q^0 - E_e^* - E_\nu^*) \frac{|k^*|^2 dk^* d\Omega^*}{E_e^* E_\nu^*}, \quad \text{where } E_e^* = \sqrt{|k^*|^2 + m_e^2}, \quad E_\nu^* = |k^*| \quad (93)$$

$$= \frac{1}{16\pi^2} \int \delta(q^0 - E_e^* - E_\nu^*) \frac{k^* dk^* d\Omega^*}{E_e^*} \quad (94)$$

$$= \frac{1}{16\pi^2} \int \delta(q^0 - \sqrt{k^{*2} + m_e^2} - k^*) \frac{k^* dk^* d\Omega^*}{E_e^*} \quad (95)$$

$$= \frac{1}{16\pi^2} \frac{k^*}{q^0} \int d\Omega^* \quad \text{with } k^* = \frac{q^2 - m_e^2}{2\sqrt{q^2}}, q^0 = \sqrt{q^2} \quad (96)$$

$$= \frac{1}{16\pi^2} \frac{k^*}{\sqrt{q^2}} \int d\Omega^* \quad (97)$$

Hence, we have

$$d\tau_2(q \rightarrow k + k') = \frac{1}{16\pi^2} \frac{q^2 - m_e^2}{2q^2} d\Omega^* = \frac{1}{16\pi^2} \frac{k^*}{\sqrt{q^2}} \int d\Omega^*. \quad (98)$$

By Lorentz transformation, we have

$$E_e = \gamma(E_e^* + \beta k^* \cos \theta^*), \quad (99)$$

$$\implies dE_e = \gamma \beta k^* d\cos \theta^* \quad (100)$$

$$\implies d\Omega^* = 2\pi d\cos \theta^* = \frac{2\pi}{\gamma \beta k^*} dE_e \quad (101)$$

$$\implies d\Omega^* = 2\pi dE_e \frac{1}{\frac{q^0}{\sqrt{q^2}} \frac{|\mathbf{q}|}{q^0} k^*}, \quad \text{where } \gamma = \frac{q^0}{\sqrt{q^2}}, \quad \beta = \frac{|\mathbf{q}|}{q^0} \quad (102)$$

$$\implies d\Omega^* = 2\pi dE_e \frac{\sqrt{q^2}}{|\mathbf{q}| k^*} \approx 2\pi dE_e \frac{\sqrt{q^2}}{\sqrt{\Delta^2 - q^2} k^*} \quad (103)$$

$$\implies d\tau_2(q \rightarrow k + k') = \frac{1}{16\pi^2} \frac{k^*}{\sqrt{q^2}} 2\pi dE_e \frac{\sqrt{q^2}}{\sqrt{\Delta^2 - q^2} k^*} \quad (104)$$

$$= \frac{1}{8\pi} \frac{1}{\sqrt{\Delta^2 - q^2}} dE_e. \quad (105)$$

Hence, the three-body phase space element and squared amplitude can be expressed as

$$d\tau_3 = d\tau_2(p \rightarrow p' + q) \frac{dq^2}{2\pi} d\tau_2(q \rightarrow k + k') = \frac{1}{4\pi} \frac{\sqrt{\Delta^2 - q^2}}{m_{\pi^-}} \frac{dq^2}{2\pi} \frac{1}{8\pi} \frac{1}{\sqrt{\Delta^2 - q^2}} dE_e \quad (106)$$

$$= \frac{1}{64\pi^3 m_{\pi^-}} dq^2 dE_e, \quad (107)$$

$$\langle |\mathcal{M}|^2 \rangle = 32G_F^2 |V_{ud}|^2 m_{\pi^-}^2 \left[2E_e E_\nu - \frac{q^2 - m_e^2}{2} - \frac{m_e^2}{m_{\pi^-}} E_\nu + \frac{m_e^2}{4m_{\pi^-}^2} \frac{q^2 - m_e^2}{2} \right] \quad (108)$$

$$\approx 32G_F^2 |V_{ud}|^2 m_{\pi^-}^2 \left[2E_e(\Delta - E_e) - \frac{q^2 - m_e^2}{2} - \frac{m_e^2}{m_{\pi^-}}(\Delta - E_e) + \frac{m_e^2}{4m_{\pi^-}^2} \frac{q^2 - m_e^2}{2} \right] \quad (109)$$

$$= 32G_F^2 |V_{ud}|^2 m_{\pi^-}^2 \left[(\Delta - E_e) \left(2E_e - \frac{m_e^2}{m_{\pi^-}} \right) - \left(1 - \frac{m_e^2}{4m_{\pi^-}^2} \right) \frac{q^2 - m_e^2}{2} \right]. \quad (110)$$

Here, we have used the approximation $E_\nu \approx m_{\pi^-} - E_e - E_{\pi^0} \approx \Delta - E_e$. Now, we can do dq^2 integration finally, and the limits of q^2 can be determined by the kinematics of the decay. $q^2 = (k + k')^2 = m_e^2 + 2E_e E_\nu(1 - \beta_e \cos \theta)$, where θ is the angle between the electron and neutrino momenta. Since $\cos \theta$ can vary from -1 to 1, we can determine the limits of q^2 by considering the extreme cases of $\cos \theta$:

$$q_{\min}^2 = m_e^2 + 2E_e E_\nu(1 - \beta_e) = m_e^2 + 2E_e E_\nu(1 - p_e/E_e), \quad (111)$$

$$q_{\max}^2 = m_e^2 + 2E_e E_\nu(1 + \beta_e) = m_e^2 + 2E_e E_\nu(1 + p_e/E_e). \quad (112)$$

$$\int_{q_{\min}^2}^{q_{\max}^2} \left[(\Delta - E_e) \left(2E_e - \frac{m_e^2}{m_{\pi^-}} \right) - \left(1 - \frac{m_e^2}{4m_{\pi^-}^2} \right) \frac{q^2 - m_e^2}{2} \right] dq^2 \quad (113)$$

$$= \left[(\Delta - E_e) \left(2E_e - \frac{m_e^2}{m_{\pi^-}} \right) (q^2 - m_e^2) - \left(1 - \frac{m_e^2}{4m_{\pi^-}^2} \right) \frac{(q^2 - m_e^2)^2}{4} \right]_{q_{\min}^2}^{q_{\max}^2} \quad (114)$$

$$= \left[(\Delta - E_e) \left(2E_e - \frac{m_e^2}{m_{\pi^-}} \right) (4p_e E_\nu) - \left(1 - \frac{m_e^2}{4m_{\pi^-}^2} \right) \frac{(4p_e E_\nu)^2}{4} \right] \quad (115)$$

$$= 4p_e E_\nu \left[(\Delta - E_e) \left(2E_e - \frac{m_e^2}{m_{\pi^-}} \right) - \left(1 - \frac{m_e^2}{4m_{\pi^-}^2} \right) p_e E_\nu \right] \quad (116)$$

$$\approx 4p_e (\Delta - E_e)^2 \left[\left(2E_e - \frac{m_e^2}{m_{\pi^-}} \right) - \left(1 - \frac{m_e^2}{4m_{\pi^-}^2} \right) p_e \right] \quad (117)$$

Finally, the decay rate can be expressed as

$$\frac{d\Gamma}{dE_e} = \frac{1}{2m_{\pi^-}} \langle |\mathcal{M}|^2 \rangle d\tau_3 \quad (118)$$

$$= \frac{1}{2m_{\pi^-}} 32G_F^2 |V_{ud}|^2 m_{\pi^-}^2 \frac{1}{64\pi^3 m_{\pi^-}} 4p_e (\Delta - E_e)^2 \left[\left(2E_e - \frac{m_e^2}{m_{\pi^-}} \right) - \left(1 - \frac{m_e^2}{4m_{\pi^-}^2} \right) p_e \right] \quad (119)$$

$$= \frac{G_F^2 |V_{ud}|^2}{\pi^3} (\Delta - E_e)^2 p_e \left[\left(2E_e - \frac{m_e^2}{m_{\pi^-}} \right) - \left(1 - \frac{m_e^2}{4m_{\pi^-}^2} \right) p_e \right] \quad (120)$$

LAST, we can take the limit $m_e \rightarrow 0$ to get the final expression for the decay rate:

$$\frac{d\Gamma}{dE_e} = \frac{G_F^2 |V_{ud}|^2}{\pi^3} (\Delta - E_e)^2 E_e \left[2E_e - E_e \right] \quad (121)$$

$$= \frac{G_F^2 |V_{ud}|^2}{\pi^3} (\Delta - E_e)^2 E_e^2. \quad (122)$$

(b)

Now, we can integrate over the electron energy E_e to get the total decay rate, the limits of E_e can be determined by the kinematics of the decay. The minimum value of E_e occurs when the electron is at rest, which gives $E_{e,\min} = m_e$. The maximum value of E_e occurs when the electron gets all the rest of energy, which gives $E_{e,\max} = \Delta$. Therefore, we have

$$\Gamma = \frac{G_F^2 |V_{ud}|^2}{\pi^3} \int_{m_e}^{\Delta} (\Delta - E_e)^2 p_e \left[\left(2E_e - \frac{m_e^2}{m_{\pi^-}} \right) - \left(1 - \frac{m_e^2}{4m_{\pi^-}^2} \right) p_e \right] dE_e \quad (123)$$

$$= \frac{G_F^2 |V_{ud}|^2}{\pi^3} \int_{m_e}^{\Delta} (\Delta - E_e)^2 \sqrt{E_e^2 - m_e^2} \left[\left(2E_e - \frac{m_e^2}{m_{\pi^-}} \right) - \left(1 - \frac{m_e^2}{4m_{\pi^-}^2} \right) \sqrt{E_e^2 - m_e^2} \right] dE_e \quad (124)$$

$$= \frac{G_F^2 |V_{ud}|^2 \Delta^5}{\pi^3} \int_{m_e/\Delta}^1 (1-x)^2 \sqrt{x^2 - (m_e/\Delta)^2} \left[\left(2x - \frac{m_e^2}{m_{\pi^-} \Delta} \right) - \left(1 - \frac{m_e^2}{4m_{\pi^-}^2} \right) \sqrt{x^2 - (m_e/\Delta)^2} \right] dx \quad (125)$$

$$\approx \frac{G_F^2 |V_{ud}|^2 \Delta^5}{\pi^3} \left(1 - \frac{3\Delta}{2m_{\pi^-}} \right) \left(\sqrt{1 - \frac{m_e^2}{\Delta^2}} \left(1 - \frac{9m_e^2}{2\Delta^2} - \frac{15m_e^4}{8\Delta^4} \right) + \frac{15m_e^4}{8\Delta^4} \ln \left[\frac{\Delta}{m_e} \left(1 + \sqrt{1 - \frac{m_e^2}{\Delta^2}} \right) \right] \right) \quad (126)$$

If we just take the leading order in m_e/Δ , we can further simplify the expression for the decay rate as

$$\Gamma \approx \frac{G_F^2 |V_{ud}|^2 \Delta^5}{\pi^3} \left(1 - \frac{3\Delta}{2m_{\pi^-}} \right) \left(\left(1 - \frac{m_e^2}{2\Delta^2} \right) \left(1 - \frac{9m_e^2}{2\Delta^2} \right) \right) \quad (127)$$

$$\approx \frac{G_F^2 |V_{ud}|^2 \Delta^5}{\pi^3} \left(1 - \frac{3\Delta}{2m_{\pi^-}} \right) \left(1 - \frac{5m_e^2}{\Delta^2} \right) \quad (128)$$

$$\approx \frac{G_F^2 |V_{ud}|^2 \Delta^5}{\pi^3} \left(1 - \frac{3\Delta}{2m_{\pi^-}} - \frac{5m_e^2}{\Delta^2} \right) \quad (129)$$

Hence, we get $a = 3/2$, $b = 5$.

□

Question 2

Tau decays

- (a) Find the decay rate for the two-body decay $\tau^- \rightarrow \pi^- + \nu_\tau$, neglecting neutrino masses and using $\langle 0 | \bar{d} \gamma^\mu \gamma_5 u | \pi^- \rangle = i f_\pi p_\pi^\mu$. Determine the ratio

$$R_\pi = \frac{\Gamma(\tau^- \rightarrow \pi^- + \nu_\tau)}{\Gamma(\tau^- \rightarrow e^- + \bar{\nu}_e + \nu_\tau)}, \quad (130)$$

using the tree-level leptonic rate with $m_e = 0$, and compare with the corresponding PDG branching-fraction ratio.

- (b) Now consider $\tau^- \rightarrow \rho^- + \nu_\tau$ with $\langle 0 | \bar{d} \gamma^\mu u | \rho^-(q, \epsilon) \rangle = f_\rho m_\rho \epsilon^\mu$, and derive the decay rate $\Gamma(\tau^- \rightarrow \rho^- + \nu_\tau)$ (neglect the neutrino mass). Form the ratio

$$R = \frac{\Gamma(\tau^- \rightarrow \rho^- + \nu_\tau)}{\Gamma(\tau^- \rightarrow e^- + \bar{\nu}_e + \nu_\tau)}, \quad (131)$$

and compare with the PDG data to extract f_ρ (or the ratio f_ρ/f_π).

Hint: Use the polarization sum

$$\sum_\lambda \epsilon_\mu^{(\lambda)}(q) \epsilon_\nu^{(\lambda)*}(q) = -g_{\mu\nu} + \frac{q_\mu q_\nu}{m_\rho^2}. \quad (132)$$

Answer

(a)

First, we can write down the amplitude for the decay $\tau^- \rightarrow \pi^- + \nu_\tau$ as

$$\mathcal{M} = \frac{G_F}{\sqrt{2}} V_{ud} \langle \pi^- | \bar{d} \gamma^\mu (1 - \gamma_5) u | 0 \rangle \langle \nu_\tau | \bar{\nu}_\tau \gamma_\mu (1 - \gamma_5) \tau^- | \tau^- \rangle \quad (133)$$

$$= i \frac{G_F}{\sqrt{2}} V_{ud} f_\pi p_\pi^\mu \bar{u}_{\nu_\tau} \gamma_\mu (1 - \gamma_5) u_\tau, \quad (134)$$

where we have used the given matrix element $\langle 0 | \bar{d} \gamma^\mu \gamma_5 u | \pi^- \rangle = i f_\pi p_\pi^\mu$ and the fact that $\langle \nu_\tau | \bar{\nu}_\tau \gamma_\mu (1 - \gamma_5) \tau^- | \tau^- \rangle$ can be expressed as $\bar{u}_{\nu_\tau} \gamma_\mu (1 - \gamma_5) u_\tau$. The average squared amplitude can be calculated as (this factor of 1/2 is due to the average over the initial τ spin states and the trace calculation is done

in the previous question):

$$|\mathcal{M}|^2 = \frac{1}{2} \frac{G_F^2}{2} |V_{ud}|^2 f_\pi^2 p_\pi^\mu p_\pi^\nu \text{Tr}[(\not{p}_{\nu_\tau}) \gamma_\mu (1 - \gamma_5) (\not{p}_\tau + m_\tau) \gamma_\nu (1 - \gamma_5)] \quad (135)$$

$$= 2G_F^2 |V_{ud}|^2 f_\pi^2 \left[2(p_\pi \cdot p_{\nu_\tau})(p_\pi \cdot p_\tau) - m_\tau^2 (p_\pi \cdot p_{\nu_\tau}) \right] \quad (136)$$

$$= G_F^2 |V_{ud}|^2 f_\pi^2 m_\tau^2 (m_\tau^2 - m_\pi^2), \quad (137)$$

where we have used the fact that $p_\pi \cdot p_{\nu_\tau} = \frac{m_\tau^2 - m_\pi^2}{2}$ and $p_\pi \cdot p_\tau = \frac{m_\tau^2 + m_\pi^2}{2}$ in the rest frame of the τ lepton. The decay rate can be calculated using the standard formula for two-body decays:

$$\Gamma = \frac{1}{2m_\tau} \int d\tau_2 |\mathcal{M}|^2 \quad (138)$$

$$= \frac{1}{2m_\tau} G_F^2 |V_{ud}|^2 f_\pi^2 m_\tau^2 (m_\tau^2 - m_\pi^2) \frac{|\mathbf{p}_\pi|}{4\pi m_\tau} \quad (139)$$

$$= \frac{G_F^2 |V_{ud}|^2 f_\pi^2 (m_\tau^2 - m_\pi^2)^2}{16\pi m_\tau}, \quad \text{where } |\mathbf{p}_\pi| = \frac{m_\tau^2 - m_\pi^2}{2m_\tau}. \quad (140)$$

Next, we can quote the result for the decay rate of $\tau^- \rightarrow e^- + \bar{\nu}_e + \nu_\tau$, which we have already calculated in the previous question:

$$\Gamma(\tau^- \rightarrow e^- + \bar{\nu}_e + \nu_\tau) = \frac{G_F^2 m_\tau^5}{192\pi^3} \left[1 - 8 \frac{m_e^2}{m_\tau^2} + \mathcal{O}\left(\frac{m_e^4}{m_\tau^4}\right) \right] \approx \frac{G_F^2 m_\tau^5}{192\pi^3}. \quad (141)$$

Now, we can form the ratio R_π as

$$R_\pi = \frac{\Gamma(\tau^- \rightarrow \pi^- + \nu_\tau)}{\Gamma(\tau^- \rightarrow e^- + \bar{\nu}_e + \nu_\tau)} \quad (142)$$

$$= \frac{\frac{G_F^2 |V_{ud}|^2 f_\pi^2 (m_\tau^2 - m_\pi^2)^2}{16\pi m_\tau}}{\frac{G_F^2 m_\tau^5}{192\pi^3}} \quad (143)$$

$$= \frac{12\pi^2 |V_{ud}|^2 f_\pi^2 (m_\tau^2 - m_\pi^2)^2}{m_\tau^6}. \quad (144)$$

Now the value $G_F = 1.166 \times 10^{-5} \text{ GeV}^{-2}$, $|V_{ud}| = 0.974$, $f_\pi = 130 \text{ MeV}$, $m_\tau = 1.777 \text{ GeV}$, and $m_\pi = 0.140 \text{ GeV}$, we can get the numerical value for R_π as

$$R_\pi = \frac{12\pi^2 |V_{ud}|^2 f_\pi^2 (m_\tau^2 - m_\pi^2)^2}{m_\tau^6} \approx 0.594. \quad (145)$$

The corresponding PDG branching-fraction ratio can be calculated as

$$R_\pi^{\text{PDG}} = \frac{\text{Br}(\tau^- \rightarrow \pi^- + \nu_\tau)}{\text{Br}(\tau^- \rightarrow e^- + \bar{\nu}_e + \nu_\tau)} = \frac{10.82\%}{17.85\%} \approx 0.606. \quad (146)$$

The theoretical prediction for R_π is in good agreement with the experimental value from PDG, which

indicates that our calculation is consistent with the experimental data.

(b)

The amplitude for the decay $\tau^- \rightarrow \rho^- + \nu_\tau$ can be written as

$$\mathcal{M} = \frac{G_F}{\sqrt{2}} V_{ud} \langle \rho^- | \bar{d} \gamma^\mu u | 0 \rangle \langle \nu_\tau | \bar{\nu}_\tau \gamma_\mu (1 - \gamma_5) \tau^- | \tau^- \rangle \quad (147)$$

$$= i \frac{G_F}{\sqrt{2}} V_{ud} f_\rho m_\rho \epsilon^\mu \bar{u}_{\nu_\tau} \gamma_\mu (1 - \gamma_5) u_\tau, \quad (148)$$

where we have used the given matrix element $\langle 0 | \bar{d} \gamma^\mu u | \rho^-(q, \epsilon) \rangle = f_\rho m_\rho \epsilon^\mu$ and the fact that $\langle \nu_\tau | \bar{\nu}_\tau \gamma_\mu (1 - \gamma_5) \tau^- | \tau^- \rangle$ can be expressed as $\bar{u}_{\nu_\tau} \gamma_\mu (1 - \gamma_5) u_\tau$. The average squared amplitude can be calculated as

$$|\mathcal{M}|^2 = \frac{1}{2} \frac{G_F^2}{2} |V_{ud}|^2 f_\rho^2 m_\rho^2 (-g_{\mu\nu} + \frac{q_\mu q_\nu}{m_\rho^2}) \text{Tr}[(\not{p}_{\nu_\tau}) \gamma^\mu (1 - \gamma_5) (\not{p}_\tau + m_\tau) \gamma^\nu (1 - \gamma_5)] \quad (149)$$

$$= \frac{1}{2} \frac{G_F^2}{2} |V_{ud}|^2 f_\rho^2 m_\rho^2 (-g_{\mu\nu} + \frac{q_\mu q_\nu}{m_\rho^2}) \left[8(p_\tau^\mu p_{\nu_\tau}^\nu + p_\tau^\nu p_{\nu_\tau}^\mu - g^{\mu\nu} (p_\tau \cdot p_{\nu_\tau}) + i\epsilon^{\alpha\mu\beta\nu} p_{\tau,\alpha} p_{\nu_\tau,\beta}) \right] \quad (150)$$

$$= 2G_F^2 |V_{ud}|^2 f_\rho^2 m_\rho^2 (-g_{\mu\nu} + \frac{q_\mu q_\nu}{m_\rho^2}) \left[p_\tau^\mu p_{\nu_\tau}^\nu + p_\tau^\nu p_{\nu_\tau}^\mu - g^{\mu\nu} (p_\tau \cdot p_{\nu_\tau}) \right] \quad (151)$$

$$= 2G_F^2 |V_{ud}|^2 f_\rho^2 m_\rho^2 \left[2(p_\tau \cdot p_{\nu_\tau}) + \frac{2(p_\tau \cdot q)(p_{\nu_\tau} \cdot q)}{m_\rho^2} - \frac{q^2}{m_\rho^2} (p_\tau \cdot p_{\nu_\tau}) \right] \quad (152)$$

$$= 2G_F^2 |V_{ud}|^2 f_\rho^2 m_\rho^2 \left[2(p_\tau \cdot p_{\nu_\tau}) + \frac{2(p_\tau \cdot q)(p_{\nu_\tau} \cdot q)}{m_\rho^2} - (p_\tau \cdot p_{\nu_\tau}) \right] \quad (153)$$

$$= 2G_F^2 |V_{ud}|^2 f_\rho^2 m_\rho^2 \left[(p_\tau \cdot p_{\nu_\tau}) + \frac{2(p_\tau \cdot q)(p_{\nu_\tau} \cdot q)}{m_\rho^2} \right]. \quad (154)$$

In the rest frame of the τ lepton, we have $p_\tau = (m_\tau, \mathbf{0})$, $p_{\nu_\tau} = (E_\nu, -\mathbf{q})$, and $q = (E_\rho, \mathbf{q})$. Hence, we can have following expressions for the dot products:

$$p_\tau \cdot p_{\nu_\tau} = \frac{1}{2} (p_\tau^2 + p_{\nu_\tau}^2 - (p_\tau - p_{\nu_\tau})^2) = \frac{1}{2} (m_\tau^2 + 0 - m_\rho^2) = \frac{1}{2} (m_\tau^2 - m_\rho^2), \quad (155)$$

$$p_\tau \cdot q = \frac{1}{2} ((p_\tau + q)^2 - p_\tau^2 - q^2) = \frac{1}{2} (m_\rho^2 + m_\tau^2), \quad (156)$$

$$p_{\nu_\tau} \cdot q = \frac{1}{2} ((p_{\nu_\tau} + q)^2 - p_{\nu_\tau}^2 - q^2) = \frac{1}{2} (m_\tau^2 - m_\rho^2). \quad (157)$$

Now, we can express the average squared amplitude as

$$|\mathcal{M}|^2 = 2G_F^2 |V_{ud}|^2 f_\rho^2 m_\rho^2 \left[\frac{1}{2}(m_\tau^2 - m_\rho^2) + \frac{2 \cdot \frac{1}{2}(m_\tau^2 + m_\rho^2) \cdot \frac{1}{2}(m_\tau^2 - m_\rho^2)}{m_\rho^2} \right] \quad (158)$$

$$= 2G_F^2 |V_{ud}|^2 f_\rho^2 m_\rho^2 \left[\frac{1}{2}(m_\tau^2 - m_\rho^2) + \frac{(m_\tau^2 + m_\rho^2)(m_\tau^2 - m_\rho^2)}{2m_\rho^2} \right] \quad (159)$$

$$= G_F^2 |V_{ud}|^2 f_\rho^2 m_\rho^2 (m_\tau^2 - m_\rho^2) \left[1 + \frac{m_\tau^2 + m_\rho^2}{m_\rho^2} \right] \quad (160)$$

$$= G_F^2 |V_{ud}|^2 f_\rho^2 (m_\tau^2 - m_\rho^2)(m_\tau^2 + 2m_\rho^2). \quad (161)$$

The decay rate can be calculated using the standard formula for two-body decays:

$$\Gamma = \frac{1}{2m_\tau} \int d\tau_2 |\mathcal{M}|^2 \quad (162)$$

$$= \frac{1}{2m_\tau} G_F^2 |V_{ud}|^2 f_\rho^2 (m_\tau^2 - m_\rho^2)(m_\tau^2 + 2m_\rho^2) \frac{|\mathbf{q}|}{4\pi m_\tau} \quad (163)$$

$$= \frac{G_F^2 |V_{ud}|^2 f_\rho^2 (m_\tau^2 - m_\rho^2)^2 (m_\tau^2 + 2m_\rho^2)}{16\pi m_\tau^3}, \quad \text{where } |\mathbf{q}| = \frac{m_\tau^2 - m_\rho^2}{2m_\tau}. \quad (164)$$

$$= \frac{G_F^2 |V_{ud}|^2 f_\rho^2 (m_\tau^2 - m_\rho^2)^2 (1 + 2\frac{m_\rho^2}{m_\tau^2})}{16\pi m_\tau}. \quad (165)$$

Now, we can form the ratio R as

$$R = \frac{\Gamma(\tau^- \rightarrow \rho^- + \nu_\tau)}{\Gamma(\tau^- \rightarrow e^- + \bar{\nu}_e + \nu_\tau)} \quad (166)$$

$$= \frac{\frac{G_F^2 |V_{ud}|^2 f_\rho^2 (m_\tau^2 - m_\rho^2)^2 (1 + 2\frac{m_\rho^2}{m_\tau^2})}{16\pi m_\tau}}{\frac{G_F^2 m_\tau^5}{192\pi^3}} \quad (167)$$

$$= \frac{12\pi^2 |V_{ud}|^2 f_\rho^2 (m_\tau^2 - m_\rho^2)^2 (1 + 2\frac{m_\rho^2}{m_\tau^2})}{m_\tau^6}. \quad (168)$$

Now the value $G_F = 1.166 \times 10^{-5} \text{ GeV}^{-2}$, $|V_{ud}| = 0.974$, $m_\tau = 1.777 \text{ GeV}$, and $m_\rho = 0.775 \text{ GeV}$, we can get the numerical value for R as

$$R = \frac{12\pi^2 |V_{ud}|^2 f_\rho^2 (m_\tau^2 - m_\rho^2)^2 (1 + 2\frac{m_\rho^2}{m_\tau^2})}{m_\tau^6} \approx 47.51 f_\rho^2. \quad (169)$$

The corresponding PDG branching-fraction ratio can be calculated as

$$R^{\text{PDG}} = \frac{\text{Br}(\tau^- \rightarrow \rho^- + \nu_\tau)}{\text{Br}(\tau^- \rightarrow e^- + \bar{\nu}_e + \nu_\tau)} = \frac{25.49\%}{17.85\%} \approx 1.428. \quad (170)$$

By comparing the theoretical prediction for R with the experimental value from PDG, we can extract

the value of f_ρ as

$$47.51 f_\rho^2 = 1.428 \tag{171}$$

$$\implies f_\rho \approx 0.173 \text{ GeV}. \tag{172}$$

Finally, we can also calculate the ratio f_ρ/f_π as

$$\frac{f_\rho}{f_\pi} = \frac{0.173 \text{ GeV}}{0.130 \text{ GeV}} \approx 1.33. \tag{173}$$

Remark: The f_ρ I found on the website is around 0.216 GeV, which is larger than the value I extracted from the τ decay data. However, I don't think this discrepancy is a problem, since the value of f_ρ can depend on the definition and the method of extraction.

Problem Set 3 Due 11am, Monday, March 2, 2026

Question

Neutral kaons: oscillations, CP violation and CPT invariance

Neutral kaon mixing is described by the effective Hamiltonian $H = M - \frac{i}{2}\Gamma$, acting on the $(K^0, \bar{K}^0)$ basis, where $M(\Gamma)$ is the neutral kaon mass (decay) matrix.

(a)

Consider the following CP-conserving parameterization of the mass matrix as defined in the $(K^0, \bar{K}^0)$ basis:

$$M_0 = \begin{pmatrix} m & \Delta \\ \Delta & m \end{pmatrix}, \quad (174)$$

where Δ is real-valued. Determine the mass eigenvalues, $m_{\pm}$ and basis states (K_-, K_+) in which M_0 becomes diagonal. Using the experimental inputs m_{K^0} and $\Delta m = m_L - m_S$, obtain numerical values for m and Δ .

(b)

Working in the (K_-, K_+) basis, extend the model of (a) to allow for CP violation by introducing the real-valued parameter δ :

$$M_{\text{CP}} = \begin{pmatrix} m_- & -i\delta \\ i\delta & m_+ \end{pmatrix}, \quad (175)$$

and assume there is no direct CP violation. This mass matrix corresponds to the *superweak* model. Assume Γ is diagonal in the (K_-, K_+) basis with eigenvalues Γ_L and Γ_S . By expressing M_{CP} in the $(K^0, \bar{K}^0)$ basis, predict the phase, φ_{ϵ} of ϵ defined by $|K_S\rangle \propto |K_+\rangle + \epsilon|K_-\rangle$ and $|K_L\rangle \propto |K_-\rangle + \epsilon|K_+\rangle$, and determine δ from the measured value of $|\epsilon|$.

(c)

Assume the $\Delta S = \Delta Q$ rule so that, to an excellent approximation, $K^0 \rightarrow \pi^- l^+ \nu$ and $\bar{K}^0 \rightarrow \pi^+ l^- \bar{\nu}$, with equal magnitudes of the corresponding decay amplitudes (no direct CP violation in semileptonic decays). Starting from an initially pure $|K^0\rangle$ beam at $t = 0$, use the time evolution in terms of $|K_S\rangle$ and $|K_L\rangle$ (including the mixing parameter ϵ defined in (b)) to derive the time-dependent semileptonic charge asymmetry

$$A(t) \equiv \frac{N_+(t) - N_-(t)}{N_+(t) + N_-(t)}, \quad (176)$$

where $N_+(t)$ and $N_-(t)$ are the number of semileptonic decays in a small time bin around proper time t that produce l^+ and l^- , respectively. Express $A(t)$ in terms of ϵ , Δm and $\Gamma_{L,S}$ (working to first order in ϵ). Show that at late times (when the beam is K_L dominated), $A(t) \rightarrow \delta_L \approx 2\text{Re}(\epsilon)$. Using PDG values for Δm , $\Gamma_{L,S}$, make a plot of $A(t)$ versus t/τ_S over the range $0 \leq t/\tau_S \leq 10$.

(d)

Finally, extend the model in (b) to

$$M_{\text{CPT}} = \begin{pmatrix} m_- & \chi \\ \chi^* & m_+ \end{pmatrix}, \quad (177)$$

where χ is complex and $\text{Re}(\chi)$ is a T -conserving, CP-violating and CPT-violating parameter. Show that, to first order in χ , the states which diagonalize the full Hamiltonian H are

$$|K_S\rangle \approx |K_+\rangle - \frac{\chi}{\xi}|K_-\rangle, \quad (178)$$

$$|K_L\rangle \approx |K_-\rangle + \frac{\chi^*}{\xi}|K_+\rangle, \quad (179)$$

where $\xi = (\Delta m)/2 - i(\Delta\Gamma)/4 \approx (m_L - m_S)/2 + i(\Gamma_S)/4$, (assuming $\Delta\Gamma \equiv \Gamma_L - \Gamma_S$ and $\Gamma_S \gg \Gamma_L$). Then include direct CP violation by writing $\eta_{+-} \approx \epsilon + \epsilon'$ and $\eta_{00} \approx \epsilon - 2\epsilon'$ (to first order in small parameters), and determine how the CPT-violating mixing term χ/ξ modifies the phases of η_{+-} and η_{00} . Derive the following relation between phases

$$\frac{2}{3}\varphi_{+-} + \frac{1}{3}\varphi_{00} - \varphi_\epsilon \approx \frac{|m_{\bar{K}^0} - m_{K^0}|}{2|\epsilon|(m_L - m_S)} \sin \varphi_\epsilon. \quad (180)$$

The result $|m_{\bar{K}^0} - m_{K^0}|/m_{K^0} < 8.0 \times 10^{-19}$, which follows from this relation, provides one of the best limits on CPT invariance.

Answer

(a)

The determinant condition for the eigenvalues of M_0 is

$$\det(M_0 - \lambda \mathbb{1}) = 0 \implies (m - \lambda)^2 - \Delta^2 = 0 \implies \lambda = m \pm \Delta. \quad (181)$$

Thus, the mass eigenvalues are $m_\pm = m \pm \Delta$. The corresponding eigenvectors are

$$|K_-\rangle = \frac{1}{\sqrt{2}}(|K^0\rangle - |\bar{K}^0\rangle), \quad (182)$$

$$|K_+\rangle = \frac{1}{\sqrt{2}}(|K^0\rangle + |\bar{K}^0\rangle). \quad (183)$$

Using the experimental inputs $m_{K^0} \approx 497.611$ MeV and $\Delta m = m_L - m_S \approx 3.484 \times 10^{-12}$ MeV from PDG, we can identify $m_S = m - \Delta$ and $m_L = m + \Delta$. Thus,

$$m_S = m - \Delta, \quad (184)$$

$$m_L = m + \Delta. \quad (185)$$

Adding these two equations gives $m_S + m_L = 2m \implies m = \frac{m_S + m_L}{2} \approx 497.611$ MeV. And $\Delta = \frac{m_L - m_S}{2} = \frac{\Delta m}{2} \approx 1.742 \times 10^{-12}$ MeV.

(b)

To express M_{CP} in the $(K^0, \bar{K}^0)$ basis, we first write the transformation from the (K_-, K_+) basis to the $(K^0, \bar{K}^0)$ basis:

$$\begin{pmatrix} |K_-\rangle \\ |K_+\rangle \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} |K^0\rangle \\ |\bar{K}^0\rangle \end{pmatrix}. \quad (186)$$

Thus, the transformation matrix is $U = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$. The mass matrix in the $(K^0, \bar{K}^0)$ basis is then given by

$$M_{\text{CP}}^{(K^0, \bar{K}^0)} = U^\dagger M_{\text{CP}} U. \quad (187)$$

Calculating this explicitly, we find

$$M_{\text{CP}}^{(K^0, \bar{K}^0)} = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} m_- & -i\delta \\ i\delta & m_+ \end{pmatrix} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} \quad (188)$$

$$= \frac{1}{2} \begin{pmatrix} m_- + i\delta & m_+ - i\delta \\ -m_- + i\delta & m_+ + i\delta \end{pmatrix} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} \quad (189)$$

$$= \frac{1}{2} \begin{pmatrix} (m_- + i\delta) + (m_+ - i\delta) & -(m_- + i\delta) + (m_+ - i\delta) \\ (-m_- + i\delta) + (m_+ + i\delta) & -(-m_- + i\delta) + (m_+ + i\delta) \end{pmatrix} \quad (190)$$

$$= \begin{pmatrix} \frac{m_- + m_+}{2} & \frac{m_+ - m_-}{2} - i\delta \\ \frac{m_+ - m_-}{2} + i\delta & \frac{m_- + m_+}{2} \end{pmatrix} \quad (191)$$

$$= \begin{pmatrix} m & \Delta - i\delta \\ \Delta + i\delta & m \end{pmatrix}. \quad (192)$$

Also, we have $\Gamma = \begin{pmatrix} \Gamma_S & 0 \\ 0 & \Gamma_L \end{pmatrix}$ in the (K_-, K_+) basis, which transforms to

$$\Gamma^{(K^0, \bar{K}^0)} = U^\dagger \Gamma U = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} \Gamma_S & 0 \\ 0 & \Gamma_L \end{pmatrix} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} \quad (193)$$

$$= \frac{1}{2} \begin{pmatrix} \Gamma_S & \Gamma_L \\ -\Gamma_S & \Gamma_L \end{pmatrix} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} \frac{\Gamma_S + \Gamma_L}{2} & \frac{\Gamma_L - \Gamma_S}{2} \\ \frac{\Gamma_L - \Gamma_S}{2} & \frac{\Gamma_S + \Gamma_L}{2} \end{pmatrix} \quad (194)$$

$$= \begin{pmatrix} \Gamma & \Gamma_{12} \\ \Gamma_{12} & \Gamma \end{pmatrix}, \quad (195)$$

where $\Gamma = \frac{\Gamma_S + \Gamma_L}{2}$ and $\Gamma_{12} = (\Gamma_L - \Gamma_S)/2$. The full Hamiltonian in the $(K^0, \bar{K}^0)$ basis is then

$$H = M_{\text{CP}}^{(K^0, \bar{K}^0)} - \frac{i}{2} \Gamma^{(K^0, \bar{K}^0)} = \begin{pmatrix} m - \frac{i}{2} \Gamma & \Delta - i\delta - \frac{i}{2} \Gamma_{12} \\ \Delta + i\delta - \frac{i}{2} \Gamma_{12} & m - \frac{i}{2} \Gamma \end{pmatrix} \quad (196)$$

$$= \begin{pmatrix} m - \frac{i}{2} \Gamma & \Delta - \frac{i}{2} \Gamma_{12} \\ \Delta - \frac{i}{2} \Gamma_{12} & m - \frac{i}{2} \Gamma \end{pmatrix} + \begin{pmatrix} 0 & -i\delta \\ i\delta & 0 \end{pmatrix} \quad (197)$$

$$= H_0 + H_{\text{CPV}}, \quad (198)$$

where H_0 is the CP-conserving part and H_{CPV} is the CP-violating part. Now, we can consider perturbation theory to find the effect of H_{CPV} on the eigenstates of H_0 . The unperturbed eigenstates of H_0 are $|K_+\rangle$ and $|K_-\rangle$ with eigenvalues $m_S - \frac{i}{2} \Gamma_S$ and $m_L - \frac{i}{2} \Gamma_L$, respectively. The perturbed eigenstates are $|K_S\rangle$ and $|K_L\rangle$. Now, we consider the perturbation theory formula for non-degenerate states, given by

$$|\psi_n^{(1)}\rangle = \sum_{m \neq n} \frac{\langle m | H' | n \rangle}{E_n^{(0)} - E_m^{(0)}} |m\rangle, \quad (199)$$

where H' is the perturbation and $E_n^{(0)}$ are the unperturbed eigenvalues. In our case, $H' = H_{\text{CPV}}$ and the unperturbed eigenvalues are $E_+ = m_S - \frac{i}{2} \Gamma_S$ and $E_- = m_L - \frac{i}{2} \Gamma_L$. Note that

$$\langle K_- | H_{\text{CPV}} | K_+ \rangle = (H_{\text{CPV}})_{-+} = (H_{\text{CPV}})_{12} = -i\delta, \quad (200)$$

$$\langle K_+ | H_{\text{CPV}} | K_- \rangle = i\delta. \quad (201)$$

Hence, the first-order correction to the $|K_+\rangle$ state is

$$|K_S\rangle \approx |K_+\rangle + \frac{\langle K_- | H_{\text{CPV}} | K_+ \rangle}{E_+ - E_-} |K_-\rangle = |K_+\rangle + \frac{-i\delta}{(m_S - m_L) - \frac{i}{2}(\Gamma_S - \Gamma_L)} |K_-\rangle \quad (202)$$

and the first-order correction to the $|K_- \rangle$ state is

$$|K_L \rangle \approx |K_- \rangle + \frac{\langle K_+ | H_{\text{CPV}} | K_- \rangle}{E_- - E_+} |K_+ \rangle = |K_- \rangle + \frac{i\delta}{(m_L - m_S) - \frac{i}{2}(\Gamma_L - \Gamma_S)} |K_+ \rangle. \quad (203)$$

We can define ϵ , which is the mixing parameter, as

$$\epsilon = \frac{\langle K_- | H_{\text{CPV}} | K_+ \rangle}{E_+ - E_-} = \frac{-i\delta}{(m_S - m_L) - \frac{i}{2}(\Gamma_S - \Gamma_L)} = \frac{-i\delta}{-\Delta m + i\Gamma_{12}} \quad (204)$$

$$= \frac{\delta}{-\Gamma_{12} - i\Delta m} = \frac{\delta}{(\Gamma_S - \Gamma_L)/2 - i(m_L - m_S)} \quad (205)$$

$$= \frac{\delta}{\sqrt{(\Gamma_S - \Gamma_L)^2/4 + (m_L - m_S)^2}} e^{i\varphi_\epsilon}, \quad (206)$$

where $\varphi_\epsilon = \tan^{-1} \left(\frac{2\Delta m}{\Gamma_S - \Gamma_L} \right)$. Using the PDG values for Δm and $\Gamma_{L,S}$, we can calculate φ_ϵ and then determine δ from the measured value of $|\epsilon|$. The PDG values are $m_L - m_S \approx 3.484 \times 10^{-12}$ MeV, $\Gamma_S \approx 7.351 \times 10^{-12}$ MeV, and $\Gamma_L \ll \Gamma_S$. Then

$$\varphi_\epsilon = \tan^{-1} \left(\frac{2\Delta m}{\Gamma_S - \Gamma_L} \right) \approx \tan^{-1} \left(\frac{2 \times 3.484 \times 10^{-12}}{7.351 \times 10^{-12}} \right) \approx 43.4678^\circ. \quad (207)$$

For the magnitude of ϵ , we have $|\epsilon| = \frac{\delta}{\sqrt{(\Gamma_S - \Gamma_L)^2/4 + (m_L - m_S)^2}}$. Using the PDG value $|\epsilon| \approx 2.228 \times 10^{-3}$, we can solve for δ :

$$\delta = |\epsilon| \sqrt{\left(\frac{\Gamma_S - \Gamma_L}{2} \right)^2 + (m_L - m_S)^2} \quad (208)$$

$$= 1.128 \times 10^{-14} \text{ MeV}. \quad (209)$$

(c)

From part (b), we have the time evolution of the states $|K_S \rangle$ and $|K_L \rangle$ in terms of $|K_+ \rangle$ and $|K_- \rangle$. The initial state at $t = 0$ is $|K^0 \rangle = \frac{1}{\sqrt{2}}(|K_S \rangle + |K_L \rangle)$. The time evolution of the state is given by

$$|K^0(t) \rangle = \frac{1}{\sqrt{2}} (e^{-i(m_S - \frac{i}{2}\Gamma_S)t} |K_S \rangle + e^{-i(m_L - \frac{i}{2}\Gamma_L)t} |K_L \rangle). \quad (210)$$

We can express $|K_S\rangle$ and $|K_L\rangle$ in terms of $|K_+\rangle$ and $|K_-\rangle$ using the results from part (b):

$$\begin{pmatrix} |K_S\rangle \\ |K_L\rangle \end{pmatrix} = \begin{pmatrix} 1 & \epsilon \\ \epsilon & 1 \end{pmatrix} \begin{pmatrix} |K_+\rangle \\ |K_-\rangle \end{pmatrix} \quad (211)$$

$$= \begin{pmatrix} 1 & \epsilon \\ \epsilon & 1 \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{2}}(|K^0\rangle + |\bar{K}^0\rangle) \\ \frac{1}{\sqrt{2}}(|K^0\rangle - |\bar{K}^0\rangle) \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & \epsilon \\ \epsilon & 1 \end{pmatrix} \begin{pmatrix} |K^0\rangle + |\bar{K}^0\rangle \\ |K^0\rangle - |\bar{K}^0\rangle \end{pmatrix} \quad (212)$$

$$= \frac{1}{\sqrt{2}} \begin{pmatrix} (1+\epsilon)|K^0\rangle + (1-\epsilon)|\bar{K}^0\rangle \\ (1+\epsilon)|K^0\rangle - (1-\epsilon)|\bar{K}^0\rangle \end{pmatrix} \quad (213)$$

Thus, the time-evolved state can be written as

$$|K^0(t)\rangle = \frac{1}{2} \left[e^{-i(m_S - \frac{i}{2}\Gamma_S)t} ((1+\epsilon)|K^0\rangle + (1-\epsilon)|\bar{K}^0\rangle) + e^{-i(m_L - \frac{i}{2}\Gamma_L)t} ((1+\epsilon)|K^0\rangle - (1-\epsilon)|\bar{K}^0\rangle) \right] \quad (214)$$

$$= \frac{1}{2} \left[(e^{-i(m_S - \frac{i}{2}\Gamma_S)t} + e^{-i(m_L - \frac{i}{2}\Gamma_L)t}) (1+\epsilon)|K^0\rangle + (e^{-i(m_S - \frac{i}{2}\Gamma_S)t} - e^{-i(m_L - \frac{i}{2}\Gamma_L)t}) (1-\epsilon)|\bar{K}^0\rangle \right] \quad (215)$$

$$= \frac{1}{2} \left[(e^{-im_S t} e^{-\frac{1}{2}\Gamma_S t} + e^{-im_L t} e^{-\frac{1}{2}\Gamma_L t}) (1+\epsilon)|K^0\rangle + (e^{-im_S t} e^{-\frac{1}{2}\Gamma_S t} - e^{-im_L t} e^{-\frac{1}{2}\Gamma_L t}) (1-\epsilon)|\bar{K}^0\rangle \right]. \quad (216)$$

The semileptonic decay amplitudes are given by

$$\langle \pi^- l^+ \nu | H | K^0 \rangle = A, \quad (217)$$

$$\langle \pi^+ l^- \bar{\nu} | H | \bar{K}^0 \rangle = A, \quad (218)$$

where A is the amplitude for the semileptonic decay. The number of decays producing l^+ and l^- at time t are proportional to the probabilities of finding $|K^0\rangle$ and $|\bar{K}^0\rangle$ in the time-evolved state, respectively. Thus,

$$N_+(t) \propto |\langle \pi^- l^+ \nu | H | K^0(t) \rangle|^2 = |A|^2 \left| \frac{1}{2} (e^{-im_S t} e^{-\frac{1}{2}\Gamma_S t} + e^{-im_L t} e^{-\frac{1}{2}\Gamma_L t}) (1+\epsilon) \right|^2, \quad (219)$$

$$= \frac{1}{4} |A|^2 |e^{-im_S t} e^{-\frac{1}{2}\Gamma_S t} + e^{-im_L t} e^{-\frac{1}{2}\Gamma_L t}|^2 (1 + 2\text{Re}(\epsilon)) = C(t)(1 + 2\text{Re}(\epsilon)), \quad (220)$$

$$N_-(t) \propto |\langle \pi^+ l^- \bar{\nu} | H | \bar{K}^0(t) \rangle|^2 = |A|^2 \left| \frac{1}{2} (e^{-im_S t} e^{-\frac{1}{2}\Gamma_S t} - e^{-im_L t} e^{-\frac{1}{2}\Gamma_L t}) (1-\epsilon) \right|^2, \quad (221)$$

$$= \frac{1}{4} |A|^2 |e^{-im_S t} e^{-\frac{1}{2}\Gamma_S t} - e^{-im_L t} e^{-\frac{1}{2}\Gamma_L t}|^2 (1 - 2\text{Re}(\epsilon)) = C'(t)(1 - 2\text{Re}(\epsilon)), \quad (222)$$

Now, we can calculate the asymmetry $A(t)$:

$$A(t) = \frac{N_+(t) - N_-(t)}{N_+(t) + N_-(t)} = \frac{C(t)(1 + 2\text{Re}(\epsilon)) - C'(t)(1 - 2\text{Re}(\epsilon))}{C(t)(1 + 2\text{Re}(\epsilon)) + C'(t)(1 - 2\text{Re}(\epsilon))} \quad (223)$$

When t is large, the beam is K_L dominated, so we can approximate $C(t) \approx C'(t)$, and thus

$$A(t) \approx \frac{(1 + 2\text{Re}(\epsilon)) - (1 - 2\text{Re}(\epsilon))}{(1 + 2\text{Re}(\epsilon)) + (1 - 2\text{Re}(\epsilon))} = \frac{4\text{Re}(\epsilon)}{2} = 2\text{Re}(\epsilon) = \delta_L. \quad (224)$$

Using the PDG values for Δm , $\Gamma_{L,S}$, and ϵ , we can plot $A(t)$ versus t/τ_S over the range $0 \leq t/\tau_S \leq 10$. The plot will show that $A(t)$ starts from zero at $t = 0$ and approaches $2\text{Re}(\epsilon)$ as t increases, reflecting the CP violation in the neutral kaon system. We can further simplify the expression for $C(t)$ and $C'(t)$ to make the plot more explicit. Note that

$$|e^{-im_S t} e^{-\frac{1}{2}\Gamma_S t} + e^{-im_L t} e^{-\frac{1}{2}\Gamma_L t}|^2 = e^{-\Gamma_S t} + e^{-\Gamma_L t} + 2e^{-\frac{1}{2}(\Gamma_S + \Gamma_L)t} \cos(\Delta m t), \quad (225)$$

$$|e^{-im_S t} e^{-\frac{1}{2}\Gamma_S t} - e^{-im_L t} e^{-\frac{1}{2}\Gamma_L t}|^2 = e^{-\Gamma_S t} + e^{-\Gamma_L t} - 2e^{-\frac{1}{2}(\Gamma_S + \Gamma_L)t} \cos(\Delta m t). \quad (226)$$

The following is the numerical parameters used for the plot:

$$\Delta m = 3.484 \times 10^{-12} \text{ MeV}, \quad (227)$$

$$\Gamma_S = 7.351 \times 10^{-12} \text{ MeV}, \quad (228)$$

$$\Gamma_L = 1.287 \times 10^{-14} \text{ MeV}, \quad (229)$$

$$|\epsilon| = 2.228 \times 10^{-3} \quad (230)$$

$$\varphi_\epsilon = 43.4678^\circ. \quad (231)$$

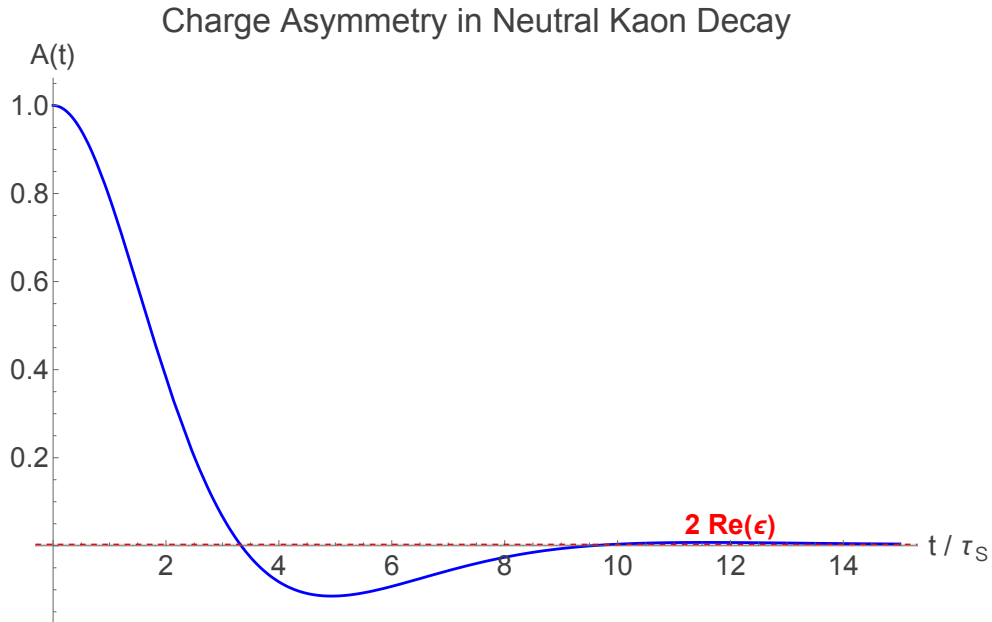


Figure 1: Plot of $A(t)$ versus t/τ_S for $0 \leq t/\tau_S \leq 15$.

(d)

To extend the model to include CPT violation, we introduce the complex parameter χ in the mass

matrix. The full Hamiltonian in the $(K^0, \bar{K}^0)$ basis is then given by

$$H = M_{\text{CPT}} - \frac{i}{2}\Gamma = \begin{pmatrix} m - \frac{i}{2}\Gamma & \Delta + \chi - \frac{i}{2}\Gamma_{12} \\ \Delta + \chi^* - \frac{i}{2}\Gamma_{12} & m - \frac{i}{2}\Gamma \end{pmatrix} \quad (232)$$

$$= \begin{pmatrix} m - \frac{i}{2}\Gamma & \Delta - \frac{i}{2}\Gamma_{12} \\ \Delta - \frac{i}{2}\Gamma_{12} & m - \frac{i}{2}\Gamma \end{pmatrix} + \begin{pmatrix} 0 & \chi \\ \chi^* & 0 \end{pmatrix} = H_0 + H_{\text{CPT}}, \quad (233)$$

To find the states that diagonalize this Hamiltonian, we can use perturbation theory again, treating χ as a small perturbation. The unperturbed eigenstates are still $|K_+\rangle$ and $|K_-\rangle$ with eigenvalues $m_S - \frac{i}{2}\Gamma_S$ and $m_L - \frac{i}{2}\Gamma_L$, respectively. The first-order correction to the $|K_+\rangle$ state is

$$|K_S\rangle \approx |K_+\rangle + \frac{\langle K_-|H_{\text{CPT}}|K_+\rangle}{E_+ - E_-}|K_-\rangle = |K_+\rangle + \frac{\chi}{(m_S - m_L) - \frac{i}{2}(\Gamma_S - \Gamma_L)}|K_-\rangle \quad (234)$$

$$\approx |K_+\rangle - \frac{\chi}{2\xi}|K_-\rangle, \quad (235)$$

and the first-order correction to the $|K_-\rangle$ state is

$$|K_L\rangle \approx |K_-\rangle + \frac{\langle K_+|H_{\text{CPT}}|K_-\rangle}{E_- - E_+}|K_+\rangle = |K_-\rangle + \frac{\chi^*}{(m_L - m_S) - \frac{i}{2}(\Gamma_L - \Gamma_S)}|K_+\rangle \quad (236)$$

$$\approx |K_-\rangle + \frac{\chi^*}{2\xi}|K_+\rangle, \quad (237)$$

where $\xi = \frac{\Delta m}{2} - i\frac{\Delta\Gamma}{4}$. **I think the equation provided in the problem statement has a typo, and the correct expressions should have a factor of 1/2 in the denominator.** Now, we can define the parameters η_{+-} and η_{00} , given by

$$\eta_{+-} = \frac{\langle \pi^+\pi^-|H|K_L\rangle}{\langle \pi^+\pi^-|H|K_S\rangle} \approx \epsilon + \epsilon' - \frac{\chi}{2\xi}, \quad (238)$$

$$\eta_{00} = \frac{\langle \pi^0\pi^0|H|K_L\rangle}{\langle \pi^0\pi^0|H|K_S\rangle} \approx \epsilon - 2\epsilon' - \frac{\chi}{2\xi}, \quad (239)$$

where ϵ is the indirect CP violation parameter and ϵ' is the direct CP violation parameter. The phases of η_{+-} and η_{00} are modified by the CPT-violating term $\chi/(2\xi)$. The phase of η_{+-} is given by

$$\varphi_{+-} = \tan^{-1} \left(\frac{\text{Im}(\epsilon + \epsilon' + \frac{\chi}{2\xi})}{\text{Re}(\epsilon + \epsilon' + \frac{\chi}{2\xi})} \right), \quad (240)$$

and the phase of η_{00} is given by

$$\varphi_{00} = \tan^{-1} \left(\frac{\text{Im}(\epsilon - 2\epsilon' + \frac{\chi}{2\xi})}{\text{Re}(\epsilon - 2\epsilon' + \frac{\chi}{2\xi})} \right). \quad (241)$$

We consider the approximation formula for the tangent of the difference of angles:

$$\text{Arg}(1+z) \approx \tan^{-1} \left(\frac{\text{Im}(z)}{1+\text{Re}(z)} \right) \approx \text{Im}(z), \quad (242)$$

where z is a small complex number. Using this approximation, we can write

$$\varphi_{+-} \approx \varphi_\epsilon + \text{Im} \left(\frac{\epsilon' + \frac{\chi}{2\xi}}{\epsilon} \right), \quad (243)$$

$$\varphi_{00} \approx \varphi_\epsilon + \text{Im} \left(\frac{-2\epsilon' + \frac{\chi}{2\xi}}{\epsilon} \right). \quad (244)$$

Now, we can derive the relation between the phases:

$$\frac{2}{3}\varphi_{+-} + \frac{1}{3}\varphi_{00} - \varphi_\epsilon \approx \frac{2}{3}\text{Im} \left(\frac{\epsilon' + \frac{\chi}{2\xi}}{\epsilon} \right) + \frac{1}{3}\text{Im} \left(\frac{-2\epsilon' + \frac{\chi}{2\xi}}{\epsilon} \right) \quad (245)$$

$$= \text{Im} \left(\frac{\chi}{2\xi\epsilon} \right) = \frac{|\chi|}{2|\xi||\epsilon|} \sin(\varphi_\chi - \varphi_\epsilon - \varphi_\xi) \approx \frac{|\chi|}{2|\xi||\epsilon|} \sin \varphi_\epsilon, \quad (246)$$

where φ_χ is the phase of χ and φ_ξ is the phase of ξ . The approximation in the last step assumes that $\varphi_\chi - \varphi_\epsilon - \varphi_\xi \approx \varphi_\epsilon$, which is reasonable if χ is small and does not introduce a large phase shift. Finally, we can express $|\chi|$ in terms of the mass difference between K^0 and $\bar{K}^0$, and ξ in terms of the mass and width differences:

$$|\chi| = |m_{\bar{K}^0} - m_{K^0}|, \quad |\xi| = \sqrt{\left(\frac{\Delta m}{2}\right)^2 + \left(\frac{\Delta \Gamma}{4}\right)^2} \approx \frac{m_L - m_S}{2}. \quad (247)$$

Thus, we have

$$\frac{2}{3}\varphi_{+-} + \frac{1}{3}\varphi_{00} - \varphi_\epsilon \approx \frac{|m_{\bar{K}^0} - m_{K^0}|}{2|\epsilon|(m_L - m_S)} \sin \varphi_\epsilon. \quad (248)$$

Now for the limit on CPT invariance, we can use the experimental values for φ_{+-} , φ_{00} , and φ_ϵ to set an upper limit on $|m_{\bar{K}^0} - m_{K^0}|$. The PDG values are $\varphi_{+-} \approx 43.4^\circ \pm 0.5^\circ$, $\varphi_{00} \approx 43.7^\circ \pm 0.6^\circ$, and $\varphi_\epsilon \approx 43.5^\circ \pm 0.5^\circ$. Using these values, we can calculate the left-hand side of the relation:

$$\frac{2}{3}\varphi_{+-} + \frac{1}{3}\varphi_{00} - \varphi_\epsilon \approx \frac{2}{3} \times 43.4^\circ + \frac{1}{3} \times 43.7^\circ - 43.5^\circ = 0^\circ \pm \sigma, \quad (249)$$

where σ is given by the error propagation formula:

$$\sigma = \sqrt{\left(\frac{2}{3}\right)^2 (0.5^\circ)^2 + \left(\frac{1}{3}\right)^2 (0.6^\circ)^2 + (0.5^\circ)^2} \approx 0.63^\circ. \quad (250)$$

Thus, we have

$$\frac{2}{3}\varphi_{+-} + \frac{1}{3}\varphi_{00} - \varphi_\epsilon = 0^\circ \pm 0.63^\circ. \quad (251)$$

Now, we can use this result to set an upper limit on $|m_{\bar{K}^0} - m_{K^0}|$:

$$\left| \frac{2}{3}\varphi_{+-} + \frac{1}{3}\varphi_{00} - \varphi_\epsilon \right| \approx \frac{|m_{\bar{K}^0} - m_{K^0}|}{2|\epsilon|(m_L - m_S)} \sin \varphi_\epsilon \leq 0.63^\circ \approx 0.011 \text{ radians}, \quad (252)$$

$$|m_{\bar{K}^0} - m_{K^0}| \leq 0.011 \times 2|\epsilon|(m_L - m_S) / \sin \varphi_\epsilon \approx 2.49 \times 10^{-16} \text{ MeV}. \quad (253)$$

Here we only consider 1σ deviation for the limit, which corresponds to a confidence level of about 68%. For a more stringent limit, we can consider 2σ or 3σ deviations, which would correspond to confidence levels of about 95% and 99.7%, respectively.

□

Problem Set 4 Due 11am, Monday, March 23, 2026

Question 1

Discrete gauge symmetry

It often occurs that spontaneous breaking of a continuous symmetry leaves a discrete subgroup unbroken. If the original symmetry was local, the remaining subgroup is known as a *discrete gauge symmetry*. As a simple example, consider a model involving three complex scalar fields $\phi_i, i = 1, 2, 3$ with a $U(1)$ gauge symmetry, $\phi_i \rightarrow e^{iq_i\alpha(x)}\phi_i$, and canonical gauge-covariant kinetic terms. Suppose the potential is of the form

$$V(\phi_1, \phi_2, \phi_3) = \sum_{i=1}^3 V_i(\phi_i^\dagger \phi_i) + V_{\text{cubic}}, \quad (254)$$

where each V_i is a renormalizable polynomial in $\phi_i^\dagger \phi_i$ (e.g. $V_i = -m_i^2 \phi_i^\dagger \phi_i + \lambda_i (\phi_i^\dagger \phi_i)^2$), and

$$V_{\text{cubic}} = \lambda_1 \phi_1 \phi_2 \phi_3 + \lambda_2 \phi_1^2 \phi_2^\dagger + \text{h.c.} \quad (255)$$

- (a) Find the $U(1)$ charge q_i of the ϕ_i for which the theory is $U(1)$ gauge invariant.
- (b) Show that by constant field rephasings $\phi_i \rightarrow e^{i\alpha_i} \phi_i$ one may take λ_1 and λ_2 to be real and positive without loss of generality.
- (c) Suppose parameters are such that the vacuum satisfies $\langle \phi_3 \rangle = v \neq 0$ but $\langle \phi_1 \rangle = \langle \phi_2 \rangle = 0$. Show that a discrete Z_3 symmetry i.e. $\phi_i \rightarrow e^{i\frac{2\pi}{3}q_i} \phi_i$ remains unbroken.

Answer

(a)

First, it is straightforward to see that the kinetic terms are already gauge invariant for any choice of q_i . Next, the potential is gauge invariant if each term is gauge invariant. The first three terms are automatically gauge invariant since they depend only on $\phi_i^\dagger \phi_i$. The last two terms are gauge invariant if

$$q_1 + q_2 + q_3 = 0, \quad (256)$$

$$2q_1 - q_2 = 0. \quad (257)$$

Solving these equations gives $q_1 = q, q_2 = 2q, q_3 = -3q$ for some arbitrary charge q .

(b)

Assume $\lambda_1 = \Lambda_1 e^{i\theta_1}$ and $\lambda_2 = \Lambda_2 e^{i\theta_2}$ with $\Lambda_{1,2} > 0$. Now, the constraint should be

$$\theta_1 + \alpha_1 + \alpha_2 + \alpha_3 = 0, \quad (258)$$

$$\theta_2 + 2\alpha_1 - \alpha_2 = 0, \quad (259)$$

where α_i are the rephasing angles. Since this is a system of two equations with three unknowns, we can always find a solution for α_i to make λ_1 and λ_2 real and positive. For example, we can choose $\alpha_1 = 0$, then $\alpha_2 = -\theta_2$ and $\alpha_3 = \theta_2 - \theta_1$. Hence, after the rephasing, we have $\lambda_1 = \Lambda_1$ and $\lambda_2 = \Lambda_2$, and both are real and positive.

(c)

Hence, if the vacuum expectation value (VEV) of ϕ_3 is nonzero, it should be invariant under the transformation. Hence

$$\langle \phi_3 \rangle \rightarrow e^{iq_3 \alpha(x)} \langle \phi_3 \rangle = e^{iq_3 \alpha(x)} v = v \quad (260)$$

$$\Rightarrow e^{iq_3 \alpha(x)} = 1 \Rightarrow q_3 \alpha(x) = 2\pi n, n \in \mathbb{Z} \Rightarrow \alpha(x) = \frac{2\pi n}{q_3} = -\frac{2\pi n}{3q}. \quad (261)$$

Hence, the corresponding transformation for ϕ_1 , ϕ_2 and ϕ_3 is

$$\phi_1 \rightarrow e^{iq_1 \alpha(x)} \phi_1 = e^{i\frac{2\pi n}{3}} \phi_1, \quad (262)$$

$$\phi_2 \rightarrow e^{iq_2 \alpha(x)} \phi_2 = e^{i\frac{4\pi n}{3}} \phi_2, \quad (263)$$

$$\phi_3 \rightarrow e^{iq_3 \alpha(x)} \phi_3 = e^{-i\frac{2\pi n}{3}} \phi_3 = \phi_3. \quad (264)$$

This is a Z_3 symmetry since $e^{i\frac{2\pi n}{3}}$ has three distinct values for $n = 0, 1, 2$ and then it repeats for $n = 3$. Therefore, the unbroken discrete gauge symmetry is Z_3 . $\square$

Question 2

Georgi-Glashow $SO(3)$ model

A competing model to what became the standard $SU(2) \times U(1)$ electroweak model was suggested by Georgi and Glashow in 1972, based on just the $SU(2) \cong SO(3)$ gauge symmetry (i.e. the symmetry with respect to three-dimensional rotations in the “weak isospin” space). The electric charge, Q was made compatible with the $SO(3)$ generators by embedding, for instance, the electron into triplets with new heavy leptons.

- (a) The $SO(3)$ symmetry is spontaneously broken by a vacuum expectation value, v of a real triplet (isovector) scalar field $\phi^a, a = 1, 2, 3$ with the Lagrangian

$$\mathcal{L} = \frac{1}{2} \partial_\mu \phi^a \partial^\mu \phi^a - \frac{\lambda}{4} (\phi^a \phi^a - v^2)^2, \quad (265)$$

where λ is a dimensionless coupling. Find the mass spectrum of the scalar particles in this model.

- (b) The $SO(3)$ symmetry is weakly gauged by adding a triplet A_μ^a gauge field with the Lagrangian

$$\mathcal{L} = -\frac{1}{4} F_{\mu\nu}^a F^{a\mu\nu} + \frac{1}{2} D_\mu \phi^a D^\mu \phi^a - \frac{\lambda}{4} (\phi^a \phi^a - v^2)^2, \quad (266)$$

where the covariant derivative, $D_\mu \phi^a = \partial_\mu \phi^a + g \varepsilon^{abc} A_\mu^b \phi^c$, and g is the gauge coupling. Find the mass spectrum of the vector and scalar particles.

- (c) Explain why this model contains no massive neutral weak boson and why this was in conflict with the 1973 observation of weak neutral currents.

Answer

- (a)

First, we can rewrite this Lagrangian with Φ as a column vector of the three components of ϕ^a :

$$\mathcal{L} = \frac{1}{2} \partial_\mu \Phi^T \partial^\mu \Phi - \frac{\lambda}{4} (\Phi^T \Phi - v^2)^2. \quad (267)$$

The vacuum expectation value (VEV) of Φ is $\langle \Phi \rangle = (0, 0, v)^T$. We can expand Φ around the VEV as

$\Phi = (h_1, h_2, v + h_3)^T$. After substituting this into the potential, we find

$$V = \frac{\lambda}{4} \left(\begin{pmatrix} h_1 \\ h_2 \\ h_3 + v \end{pmatrix} \begin{pmatrix} h_1 \\ h_2 \\ h_3 + v \end{pmatrix} - v^2 \right)^2 \quad (268)$$

$$= \frac{\lambda}{4} (h_1^2 + h_2^2 + (h_3 + v)^2 - v^2)^2 \quad (269)$$

$$= \frac{\lambda}{4} (h_1^2 + h_2^2 + h_3^2 + 2vh_3)^2 \quad (270)$$

$$= \frac{\lambda}{4} (h_1^2 + h_2^2 + h_3^2)^2 + \frac{\lambda}{2} vh_3 (h_1^2 + h_2^2 + h_3^2) + \underbrace{\lambda v^2}_{=\frac{m_{h_3}^2}{2}} h_3^2. \quad (271)$$

Hence, the mass spectrum of the scalar particles is $m_{h_1}^2 = m_{h_2}^2 = 0$ and $m_{h_3}^2 = 2\lambda v^2$. The two massless scalars h_1 and h_2 are the Goldstone bosons corresponding to the spontaneous breaking of the $SO(3)$ symmetry. Note that since h_i are real scalars, we need to divide the mass term by 2 to get the correct mass squared.

(b)

We start with the covariant derivative term:

$$D_\mu \phi^a = \partial_\mu \phi^a + g\varepsilon^{abc} A_\mu^b \phi^c. \quad (272)$$

The kinetic term for the scalar field is:

$$\frac{1}{2} D_\mu \phi^a D^\mu \phi^a = \frac{1}{2} (\partial_\mu \phi^a + g\varepsilon^{abc} A_\mu^b \phi^c) (\partial^\mu \phi^a + g\varepsilon^{ade} A^{\mu d} \phi^e) \quad (273)$$

$$= \frac{1}{2} \partial_\mu \phi^a \partial^\mu \phi^a + \frac{1}{2} g^2 \varepsilon^{abc} \varepsilon^{ade} A_\mu^b A^{\mu d} \phi^c \phi^e + g\varepsilon^{abc} A_\mu^b \phi^c \partial^\mu \phi^a. \quad (274)$$

Here, we can apply the identity $\varepsilon^{abc} \varepsilon^{ade} = \delta^{bd} \delta^{ce} - \delta^{be} \delta^{cd}$ to simplify the second term:

$$\frac{1}{2} g^2 \varepsilon^{abc} \varepsilon^{ade} A_\mu^b A^{\mu d} \phi^c \phi^e = \frac{1}{2} g^2 (\delta^{bd} \delta^{ce} - \delta^{be} \delta^{cd}) A_\mu^b A^{\mu d} \phi^c \phi^e \quad (275)$$

$$= \frac{1}{2} g^2 A_\mu^b A^{\mu b} \phi^c \phi^c - \frac{1}{2} g^2 A_\mu^b A^{\mu d} \phi^b \phi^d. \quad (276)$$

We can write down the first term and the second term explicitly as (with $(\phi^1, \phi^2, \phi^3) = (h_1, h_2, v + h_3)$):

$$\frac{1}{2} g^2 A_\mu^b A^{\mu b} \phi^c \phi^c \quad (277)$$

$$= \frac{1}{2} g^2 (A_\mu^1 A^{\mu 1} + A_\mu^2 A^{\mu 2} + A_\mu^3 A^{\mu 3}) (h_1^2 + h_2^2 + (v + h_3)^2) \quad (278)$$

$$= \frac{1}{2} g^2 (A_\mu^1 A^{\mu 1} + A_\mu^2 A^{\mu 2} + A_\mu^3 A^{\mu 3}) (h_1^2 + h_2^2 + h_3^2 + 2vh_3 + v^2) \quad (279)$$

$$\implies \frac{1}{2} g^2 v^2 (A_\mu^1 A^{\mu 1} + A_\mu^2 A^{\mu 2} + A_\mu^3 A^{\mu 3}) + \dots, \quad (280)$$

and

$$\frac{1}{2}g^2 A_\mu^b A^{\mu d} \phi^b \phi^d = \frac{1}{2}g^2 A_\mu^b \phi^b A^{\mu d} \phi^d \quad (281)$$

$$= \frac{1}{2}g^2 (A_\mu^1 h_1 + A_\mu^2 h_2 + A_\mu^3 (v + h_3)) (A^{\mu 1} h_1 + A^{\mu 2} h_2 + A^{\mu 3} (v + h_3)) \quad (282)$$

$$= \frac{1}{2}g^2 (A_\mu^1 h_1 + A_\mu^2 h_2 + A_\mu^3 (v + h_3)) (A^{\mu 1} h_1 + A^{\mu 2} h_2 + A^{\mu 3} (v + h_3)) \quad (283)$$

$$\implies \frac{1}{2}g^2 v^2 A_\mu^3 A^{\mu 3} + \dots \quad (284)$$

Combining these two terms, we find the mass term for the gauge bosons:

$$\frac{1}{2}g^2 v^2 (A_\mu^1 A^{\mu 1} + A_\mu^2 A^{\mu 2} + A_\mu^3 A^{\mu 3}) - \frac{1}{2}g^2 v^2 A_\mu^3 A^{\mu 3} \quad (285)$$

$$= \frac{1}{2}g^2 v^2 (A_\mu^1 A^{\mu 1} + A_\mu^2 A^{\mu 2}) \quad (286)$$

$$= \frac{1}{2}m_A^2 (A_\mu^1 A^{\mu 1} + A_\mu^2 A^{\mu 2}), \quad (287)$$

where $m_A^2 = g^2 v^2$. Hence, the mass spectrum of the vector bosons is $m_{A^1} = m_{A^2} = gv$ and $m_{A^3} = 0$. The mass spectrum of the scalar particles is the same as in part (a), with $m_{h_1}^2 = m_{h_2}^2 = 0$ and $m_{h_3}^2 = 2\lambda v^2$. The two massless scalars h_1 and h_2 are the Goldstone bosons corresponding to the spontaneous breaking of the $SO(3)$ symmetry, and they are "eaten" by the gauge bosons A_μ^1 and A_μ^2 to become their longitudinal components, giving them mass. The remaining scalar h_3 is a physical Higgs-like boson. In summary, the mass spectrum is:

$$m_{A^1} = m_{A^2} = gv, \quad m_{A^3} = 0, \quad (288)$$

$$m_{h_1}^2 = m_{h_2}^2 = 0, \quad m_{h_3} = \sqrt{2\lambda v^2}. \quad (289)$$

(c)

In this model, there are two massive gauge bosons, one massless gauge boson, and one physical scalar boson. The massless gauge boson A_μ^3 corresponds to an unbroken $U(1)$ symmetry, which can be identified with electromagnetism. The two massive gauge bosons A_μ^1 and A_μ^2 correspond to the charged weak bosons $W^\pm$, but there is **no neutral weak boson** like the Z boson in the Standard Model. This is the main reason why the Georgi-Glashow $SO(3)$ model was ruled out by the 1973 observation of weak neutral currents. $\square$

Question 3

Longitudinal WZ scattering and the role of the Higgs

Consider tree-level scattering $W_L^+(p_1)Z_L(p_2) \rightarrow W_L^+(p_3)Z_L(p_4)$ in the electroweak theory. Work in the custodial limit $g' \rightarrow 0$, so $m_W = m_Z = m$ and $v = 2m/g$. Assume $E \gg m$ and use

$$\varepsilon_L^\mu(p_1) = \frac{1}{m}p_1^\mu + \frac{2m}{t-2m^2}p_3^\mu + \mathcal{O}\left(\frac{m}{E}\right)^2, \quad \varepsilon_L^\mu(p_2) = \frac{1}{m}p_2^\mu + \frac{2m}{t-2m^2}p_4^\mu + \mathcal{O}\left(\frac{m}{E}\right)^2, \quad (290)$$

$$\varepsilon_L^\mu(p_3) = \frac{1}{m}p_3^\mu + \frac{2m}{t-2m^2}p_1^\mu + \mathcal{O}\left(\frac{m}{E}\right)^2, \quad \varepsilon_L^\mu(p_4) = \frac{1}{m}p_4^\mu + \frac{2m}{t-2m^2}p_2^\mu + \mathcal{O}\left(\frac{m}{E}\right)^2 \quad (291)$$

and

$$s = (p_1 + p_2)^2, \quad t = (p_1 - p_3)^2 = (p_2 - p_4)^2, \quad u = (p_1 - p_4)^2 = (p_2 - p_3)^2, \quad s + t + u = 4m^2 \approx 0. \quad (292)$$

where s, t, u are the Mandelstam variables. You may use the following Feynman rules (all momenta incoming), with Minkowski metric $\eta_{\mu\nu} = \text{diag}(+, -, -, -)$:

$$D_{\mu\nu}(k) = \frac{-i}{k^2 - m^2} \left(\eta_{\mu\nu} - \frac{k_\mu k_\nu}{m^2} \right), \quad (293)$$

$$D_h(k) = \frac{i}{k^2 - m_h^2}, \quad (294)$$

$$V^{\alpha\beta\mu}(p_+, p_-, p_Z) = -ig \left[\eta^{\alpha\beta}(p_+ - p_-)^\mu + \eta^{\beta\mu}(p_- - p_Z)^\alpha + \eta^{\mu\alpha}(p_Z - p_+)^\beta \right], \quad (295)$$

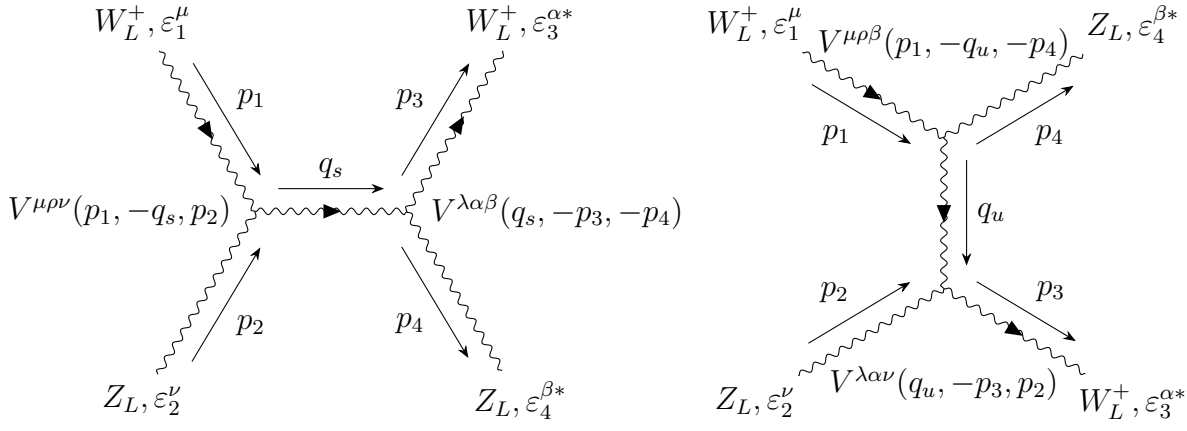
$$V^{\alpha\beta\mu\nu} = -ig^2 \left[2\eta^{\alpha\beta}\eta^{\mu\nu} - \eta^{\alpha\mu}\eta^{\beta\nu} - \eta^{\alpha\nu}\eta^{\beta\mu} \right], \quad (296)$$

$$hW_\alpha^+ W_\beta^- : V_{\alpha\beta}^{hWW} = igm\eta_{\alpha\beta}, \quad hZ_\mu Z_\nu : V_{\mu\nu}^{hZZ} = igm\eta_{\mu\nu}. \quad (297)$$

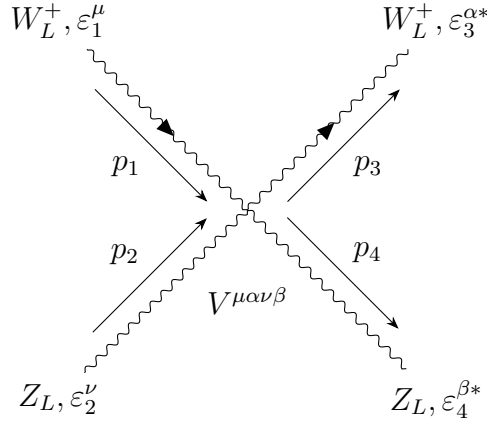
- (a) Using gauge interactions only (no Higgs exchange), draw all contributing tree-level diagrams. With $\varepsilon_L^\mu(p) \approx p^\mu/m$, show that each diagram individually contains terms of order $g^2 E^4/m^4$, but that the complete $\mathcal{O}(E^4)$ contribution cancels in the sum of all gauge diagrams. Conclude that the leading gauge-sector growth is at most $\mathcal{M}_{\text{gauge}} \propto g^2 E^2/m^2$ (up to corrections suppressed by m/E).
- (b) Now include a physical Higgs boson h of mass m_h . Draw the additional Higgs-exchange diagram(s) and determine their leading high-energy behavior using $\varepsilon_L^\mu(p) \approx p^\mu/m$. Show that adding the Higgs contribution to your result from part (a) cancels the remaining $\mathcal{O}(E^2)$ growth at fixed scattering angle, and determine the resulting constant high-energy limit of the full amplitude, expressing it in terms of the dimensionless Higgs quartic coupling λ .

Answer

(a)



(a) Feynman diagram for $W_L^+ Z_L \rightarrow W_L^+ Z_L$ via s-channel gauge boson exchange. $q_s = p_1 + p_2 = p_3 + p_4$ is the momentum of the intermediate gauge boson. (b) Feynman diagram for $W_L^+ Z_L \rightarrow W_L^+ Z_L$ via u-channel gauge boson exchange. $q_u = p_1 - p_4 = p_2 - p_3$ is the momentum of the intermediate gauge boson.



(c) Feynman diagram for $W_L^+ Z_L \rightarrow W_L^+ Z_L$ via four-point gauge interaction.

Figure 2: Tree-level Feynman diagrams for $W_L^+ Z_L \rightarrow W_L^+ Z_L$ scattering via gauge interactions.

The tree-level diagrams for $W_L^+ Z_L \rightarrow W_L^+ Z_L$ scattering via gauge interactions are shown in Figure 2a, Figure 2b and Figure 2c. We can calculate the amplitude for each diagram using the given Feynman rules and the polarization vectors. For the s-channel diagram in Figure 2a, the amplitude is:

$$i\mathcal{M}_s = V^{\mu\rho\nu}(p_1, -q_s, p_2) D_{\rho\lambda}(q_s) V^{\lambda\alpha\beta}(q_s, -p_3, -p_4) \varepsilon_{1\mu} \varepsilon_{2\nu} \varepsilon_{3\alpha}^* \varepsilon_{4\beta}^*. \quad (298)$$

Next, for the u-channel diagram in Figure 2b, the amplitude is:

$$i\mathcal{M}_u = V^{\mu\rho\beta}(p_1, -q_u, -p_4) D_{\rho\lambda}(q_u) V^{\lambda\alpha\nu}(q_u, -p_3, p_2) \varepsilon_{1\mu} \varepsilon_{2\nu} \varepsilon_{3\alpha}^* \varepsilon_{4\beta}^*. \quad (299)$$

Last, for the four-point diagram in Figure 2c, the amplitude is:

$$i\mathcal{M}_4 = V^{\mu\alpha\nu\beta} \varepsilon_{1\mu} \varepsilon_{2\nu} \varepsilon_{3\alpha}^* \varepsilon_{4\beta}^*. \quad (300)$$

All of calculation are done in Mathematica, and the results are:

$$i\mathcal{M}_s = ig^2 \frac{s^2 t^4 (s + 2t) - 8m^2 (st^4 (3s + t)) + 4m^4 t^3 (20s^2 + 21st + 2t^2) + O(m^5)}{4m^4 (m^2 - s) (t - 2m^2)^4}, \quad (301)$$

$$i\mathcal{M}_u = ig^2 \frac{-t^4 (s - t)(s + t)^2 - 4m^2 (t^4 (s + t)(3s + 5t)) + 4m^4 t^3 (20s^2 + 55st + 37t^2) + O(m^5)}{4m^4 (t - 2m^2)^4 (-3m^2 + s + t)}, \quad (302)$$

$$i\mathcal{M}_4 = ig^2 \frac{-ig^2 t^4 (-2s^2 - 2st + t^2) + 4ig^2 m^2 t^4 (3t - 2s) - 48ig^2 m^4 t^4 + O(m^5)}{4m^4 (t - 2m^2)^4}. \quad (303)$$

If we set $s = 4E^2 \gg m^2$, it is obvious that each diagram contains terms of $\frac{s^2}{m^4} \propto \frac{E^4}{m^4}$. However, when we sum up all the contributions, we find

$$i\mathcal{M}_s + i\mathcal{M}_u + i\mathcal{M}_4 = ig^2 \frac{-st^5 (s + t) + m^2 t^4 (14s^2 + 18st + 3t^2) - 8m^4 (t^3 (4s^2 + 11st + 5t^2)) + O(m^5)}{4m^2 (m^2 - s) (t - 2m^2)^4 (-3m^2 + s + t)} \quad (304)$$

$$= ig^2 \frac{-st^5 (s + t) + m^2 t^4 (14s^2 + 18st + 3t^2) - 8m^4 (t^3 (4s^2 + 11st + 5t^2)) + O(m^5)}{-4m^2 (st^4 (s + t)) + 4m^4 t^3 (8s^2 + 12st + t^2) + O(m^5)} \quad (305)$$

$$\approx ig^2 \frac{-st^5 (s + t) + m^2 t^4 (14s^2 + 18st + 3t^2) - 8m^4 (t^3 (4s^2 + 11st + 5t^2)) + O(m^5)}{-4m^2 (st^4 (s + t))} \quad (306)$$

$$\times \left(1 - \frac{4m^4 t^3 (8s^2 + 12st + t^2)}{-4m^2 (st^4 (s + t))} + O(m^5) \right) \quad (307)$$

$$\approx ig^2 \frac{-st^5 (s + t) + 2m^2 t^4 (3s^2 + 3st + t^2) + \frac{m^4 t^3 (80s^4 + 192s^3 t + 126s^2 t^2 + 14st^3 + 3t^4)}{s(s+t)} + O(m^5)}{-4m^2 (st^4 (s + t))}. \quad (308)$$

If we just check the leading term, we find this term is of order $ig^2 \frac{t}{4m^2}$. If we take $t \sim s \sim E^2$, we find the leading term is of order $ig^2 \frac{E^2}{m^2}$. Hence, the leading gauge-sector growth is at most $\mathcal{M}_{\text{gauge}} \propto g^2 E^2 / m^2$ (up to corrections suppressed by m/E).

(b)

The additional diagram for $W_L^+ Z_L \rightarrow W_L^+ Z_L$ scattering via Higgs exchange is shown in Figure 3. The amplitude for this diagram is:

$$i\mathcal{M}_h = V_{\mu\alpha}^{hWW} D_h(q_t) V_{\nu\beta}^{hZZ} \varepsilon_{1\mu} \varepsilon_{2\nu} \varepsilon_{3\alpha}^* \varepsilon_{4\beta}^*. \quad (309)$$

Again, by Mathematica, we find

$$i\mathcal{M}_h = -ig^2 \frac{t^2 (t - 4m^2)^2}{4m^4 (t - 2m^2)^2 (t - M^2)}, \quad (310)$$

where M is the mass of the Higgs boson. By estimation of order of magnitude, we find $i\mathcal{M}_h \sim -ig^2 \frac{t}{4m^2}$. Hence, the Higgs contribution cancels the remaining $\mathcal{O}(E^2)$ growth at fixed scattering angle. Now, we

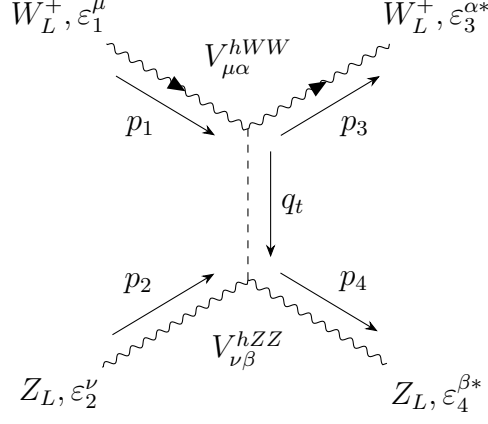


Figure 3: Feynman diagram for $W_L^+ Z_L \rightarrow W_L^+ Z_L$ scattering via Higgs exchange. $q_t = p_1 - p_3 = p_2 - p_4$ is the momentum of the intermediate Higgs boson.

can add the Higgs contribution to the gauge contribution to get the full amplitude:

$$i\mathcal{M}_{\text{full}} = i\mathcal{M}_s + i\mathcal{M}_u + i\mathcal{M}_4 + i\mathcal{M}_h \quad (311)$$

$$= ig^2 \frac{(M^2 s t^5 (s+t) + O(M^4)) + m^2 (2t^5 (s^2 + ts + t^2) + t^4 (-14s^2 - 18ts - 3t^2) M^2 + O(M^4)) + O(m^4)}{4m^2 (m^2 - s) (t - 2m^2)^4 (t - M^2) (-3m^2 + s + t)} \quad (312)$$

$$= ig^2 \frac{t^5 (M^2 s (s+t) + m^2 (2(s^2 + ts + t^2))) + \text{higher order terms}}{4m^2 (m^2 - s) (t - 2m^2)^4 (t - M^2) (-3m^2 + s + t)} \quad (313)$$

$$\approx ig^2 \frac{t^5 (M^2 s (s+t) + m^2 (2(s^2 + ts + t^2)))}{4m^2 (-s) (t^4) (t) (s+t)} \quad (314)$$

$$= -ig^2 \left(\frac{M^2}{4m^2} + \frac{2(s^2 + ts + t^2)}{s(s+t)} \right) \quad (315)$$

Here, we consider the limit $s \sim t \sim E^2 \gg m^2, M^2$. In this limit, we find the full amplitude approaches a constant:

$$i\mathcal{M}_{\text{full}} \approx -ig^2 \left(\frac{M^2}{4m^2} + \frac{2(s^2 + ts + t^2)}{s(s+t)} \right) \approx -ig^2 \left(\frac{M^2}{4m^2} + \frac{3}{2} \right). \quad (316)$$

Since $M^2 = 2\lambda v^2 = 8\lambda m^2/g^2$, we can express the constant high-energy limit of the full amplitude in terms of the dimensionless Higgs quartic coupling λ :

$$i\mathcal{M}_{\text{full}} \approx -ig^2 \left(\frac{8\lambda m^2/g^2}{4m^2} + \frac{3}{2} \right) = -ig^2 \left(2\lambda/g^2 + \frac{3}{2} \right). \quad (317)$$

□

Problem Set 5 Due 11am, Monday, April 6, 2026

Question 1

Fermion mass diagonalization for a complex mass matrix

The mass term for three chiral fermions $\psi_{L,R}^0 = (\psi_{1L,R}^0, \psi_{2L,R}^0, \psi_{3L,R}^0)$ is $\mathcal{L}_{\text{mass}} = -\overline{\psi_L^0} M \psi_R^0 + \text{h.c.}$, where

$$M = \begin{pmatrix} 1 & i & 0.5 \\ 0 & 3 + \frac{i}{2} & 2 \\ 0.25 & \frac{i}{7} & 6 \end{pmatrix}. \quad (318)$$

The physical masses are obtained from a bi-unitary transformation $A_L^\dagger M A_R = \text{diag}(m_1, m_2, m_3)$, where $m_i \geq 0$.

- (a) Show that A_L diagonalizes $MM^\dagger$, while A_R diagonalizes $M^\dagger M$ with eigenvalues m_i^2 . Explain why the physical masses are the singular values of M , rather than the eigenvalues of M .
- (b) Compute the physical masses m_1, m_2 , and m_3 , ordered as $m_1 \leq m_2 \leq m_3$, and find one pair of unitary matrices A_L and A_R such that $M = A_L \text{diag}(m_1, m_2, m_3) A_R^\dagger$.
- (c) Briefly explain why the matrices A_L and A_R are not unique, even after the masses are fixed.

Answer

(a)

First, using singular value decomposition (SVD) in Mathematica, we can find:

$$A_L = \begin{pmatrix} 0.099671 + 0.i & 0.348233 + 0.i & -0.932094 + 0.i \\ 0.371683 - 0.137417i & -0.00441727 - 0.854405i & 0.0380947 - 0.333902i \\ 0.872506 - 0.267894i & -0.0552851 + 0.381643i & 0.0726444 + 0.113936i \end{pmatrix}, \quad (319)$$

$$A_R = \begin{pmatrix} 0.0492249 - 0.0103738i & 0.111144 + 0.0317104i & -0.991542 + 0.0309028i \\ 0.156144 - 0.127386i & -0.128268 - 0.96428i & -0.0394805 - 0.107446i \\ 0.933737 - 0.291541i & -0.0553138 + 0.193115i & 0.0499158 + 0.0171554i \end{pmatrix}, \quad (320)$$

$$M_D = \begin{pmatrix} 6.45603 & 0. & 0. \\ 0. & 3.00881 & 0. \\ 0. & 0. & 0.921729 \end{pmatrix}, \quad (321)$$

such that

$$A_L^\dagger M A_R = M_D, \quad (322)$$

$$A_L^\dagger M M^\dagger A_L = M_D^2, \quad (323)$$

$$A_R^\dagger M^\dagger M A_R = M_D^2. \quad (324)$$

The reason for using singular values instead of eigenvalues is that physical masses must be real and non-negative. For a general complex mass matrix M , the eigenvalues can be complex. However, the singular values (which are the square roots of the eigenvalues of the positive-semidefinite matrix $M^\dagger M$) are guaranteed to be real and non-negative.

(b)

In order to get $m_1 \leq m_2 \leq m_3$, we can switch the first column vector and the third column vector in A_L and A_R matrices, given by

$$A'_L = \begin{pmatrix} -0.932094 + 0.i & 0.348233 + 0.i & 0.099671 + 0.i \\ 0.0380947 - 0.333902i & -0.00441727 - 0.854405i & 0.371683 - 0.137417i \\ 0.0726444 + 0.113936i & -0.0552851 + 0.381643i & 0.872506 - 0.267894i \end{pmatrix} \quad (325)$$

$$A'_R = \begin{pmatrix} -0.991542 + 0.0309028i & 0.111144 + 0.0317104i & 0.0492249 - 0.0103738i \\ -0.0394805 - 0.107446i & -0.128268 - 0.96428i & 0.156144 - 0.127386i \\ 0.0499158 + 0.0171554i & -0.0553138 + 0.193115i & 0.933737 - 0.291541i \end{pmatrix}, \quad (326)$$

such that

$$(A'_L)^\dagger M A'_R = M'_D = \begin{pmatrix} 0.921729 & 0. & 0. \\ 0. & 3.00881 & 0. \\ 0. & 0. & 6.45603 \end{pmatrix}. \quad (327)$$

Later, we just omit the prime. Now, we obtain:

$$A_L^\dagger M A_R = M_D \quad (328)$$

$$\implies A_L M_D A_R^\dagger = M. \quad (329)$$

(c)

We can define the phase angles θ_i , $i = 1, 2, 3$ such that

$$A_L \rightarrow A_L U = A_L \begin{pmatrix} e^{i\theta_1} & 0 & 0 \\ 0 & e^{i\theta_2} & 0 \\ 0 & 0 & e^{i\theta_3} \end{pmatrix}, A_R \rightarrow A_R U = A_R \begin{pmatrix} e^{i\theta_1} & 0 & 0 \\ 0 & e^{i\theta_2} & 0 \\ 0 & 0 & e^{i\theta_3} \end{pmatrix}, \quad (330)$$

and then we have

$$A_L^\dagger M A_R \rightarrow U^\dagger A_L^\dagger M A_R U = U^\dagger M_D U = M_D. \quad (331)$$

Since U and M_D are diagonal, so we can easily compute it and simplify it. In summary, A_L and A_R are not unique, by rotating with arbitrary phase angles θ_i , even though M_D is fixed the order. $\square$

Question 2

CKM-like mixing with a singlet down quark

Consider a variant of the Standard Model in which the quark sector does not contain the c, b or t quarks. The remaining quark fields consist of an $SU(2)_L$ doublet $Q_L = (u_L, d_L)^T$, together with $SU(2)_L$ singlets u_R, d_R, s_R . In addition, s_L is a left-handed $SU(2)_L$ singlet. The $SU(3)_C$ and $U(1)_Y$ quantum numbers are chosen so that all quarks have the same color and electric charges as in the Standard Model.

- Write down the charged- and neutral-current electroweak interactions, the Yukawa interactions, and any possible bare mass terms. What are the mass terms after spontaneous symmetry breaking?
- Determine the number of physical parameters in the quark sector. Separate them into masses, mixing angles and phases. Is there CP violation in this model?

Answer

Field	$SU(2)_L$ (T_3)	$U(1)_Y$	$Q = T_3 + Y$
$Q_L = (u_L, d_L)^T$	$(+1/2, -1/2)$	$+1/6$	$(+2/3, -1/3)$
u_R	0	$+2/3$	$+2/3$
d_R	0	$-1/3$	$-1/3$
s_L	0	$-1/3$	$-1/3$
s_R	0	$-1/3$	$-1/3$
$\Phi = (\phi^+, \phi^0)^T$	$(+1/2, -1/2)$	$+1/2$	$(+1, 0)$
$\tilde{\Phi} = (\phi^{0*}, -\phi^-)^T$	$(+1/2, -1/2)$	$-1/2$	$(0, -1)$

Table 1: Field quantum numbers under $SU(2)_L \times U(1)_Y$.

Table 1 shows the all quantum number in $SU(2)_L \times U(1)_Y$ for necessary quantum field. We can write down the Lagrangian, given by

$$\mathcal{L} \supset \mathcal{L}_{\text{EW}} + \mathcal{L}_{\text{Yukawa}}, \quad (332)$$

where

$$\mathcal{L}_{\text{EW}} \supset \overline{Q_L} \gamma^\mu D_\mu Q_L + \overline{u_R} \gamma^\mu D_\mu u_R + \overline{d_R} \gamma^\mu D_\mu d_R + \overline{s_L} \gamma^\mu D_\mu s_L + \overline{s_R} \gamma^\mu D_\mu s_R, \quad (333)$$

where $D_\mu = \partial_\mu - igW_\mu^a T^a - ig'B_\mu Y$. Since only Q_L has non-zero T_3 values, which involves charged-current, we can obtain:

$$\mathcal{L}_{\text{CC}} = \frac{g}{\sqrt{2}} \overline{u_L} \gamma^\mu d_L W_\mu^+ + \text{h.c.} \quad (334)$$

Here we use the standard notation for the charged gauge bosons:

$$W_\mu^\pm = \frac{1}{\sqrt{2}}(W_\mu^1 \mp iW_\mu^2). \quad (335)$$

For the neutral-current (NC) interactions, they are mediated by the photon (A_μ) and the Z boson (Z_μ), which are combinations of W_μ^3 and B_μ :

$$\mathcal{L}_{\text{NC}} = eJ_{\text{em}}^\mu A_\mu + \frac{g}{\cos \theta_w}(J_3^\mu - \sin^2 \theta_w J_{\text{em}}^\mu)Z_\mu, \quad (336)$$

where $e = g \sin \theta_w$. The electromagnetic current J_{em}^μ is summed over all fields with their respective charges Q . The critical part is the weak isospin current J_3^μ , which only receives contributions from the $SU(2)_L$ doublet Q_L :

$$J_3^\mu = \frac{1}{2}\overline{u_L}\gamma^\mu u_L - \frac{1}{2}\overline{d_L}\gamma^\mu d_L. \quad (337)$$

Note that s_L, s_R, u_R, d_R do not contribute to J_3^μ as they are singlets. Next, we write down the Yukawa interactions and the bare mass terms. Since s_L is a gauge singlet, it can form gauge-invariant mass terms directly with d_R and s_R without the Higgs field:

$$\mathcal{L}_{\text{Yukawa}} = -y_u \overline{Q_L} \tilde{\Phi} u_R - y_{d1} \overline{Q_L} \Phi d_R - y_{d2} \overline{Q_L} \Phi s_R + \text{h.c.} \quad (338)$$

$$\mathcal{L}_{\text{bare}} = -M_{s1} \overline{s_L} d_R - M_{s2} \overline{s_L} s_R + \text{h.c.} \quad (339)$$

After Spontaneous Symmetry Breaking (SSB), the Higgs field acquires a VEV $\langle \Phi \rangle = (0, v/\sqrt{2})^T$. The mass terms for the quarks are obtained by replacing Φ with its VEV:

$$\mathcal{L}_{\text{mass}} = -m_u \overline{u_L} u_R - \begin{pmatrix} \overline{d_L} & \overline{s_L} \end{pmatrix} \begin{pmatrix} y_{d1} \frac{v}{\sqrt{2}} & y_{d2} \frac{v}{\sqrt{2}} \\ M_{s1} & M_{s2} \end{pmatrix} \begin{pmatrix} d_R \\ s_R \end{pmatrix} + \text{h.c.} \quad (340)$$

$$= -m_u \overline{u_L} u_R - \begin{pmatrix} \overline{d_L} & \overline{s_L} \end{pmatrix} M_d \begin{pmatrix} d_R \\ s_R \end{pmatrix} + \text{h.c.} \quad (341)$$

where $m_u = y_u v/\sqrt{2}$. We can define the 2×2 down-type mass matrix as M_d .

(b)

Now, we have to diagonalize the matrix M_d . We can consider the two 2×2 unitary matrices U_L and U_R such that $U_L^\dagger M_d U_R$ is a diagonal matrix and all diagonal components are non-negative and real by the singular value decomposition (SVD). After rotation, there are three physical quark masses. Before rotation, there are totally 9 parameters in the $\mathcal{L}_{\text{mass}}$: 1 in m_u , 8 in M_d (4 complex numbers = 8 real numbers).

For U_R , since d_R and s_R have the same quantum number, there is no change in the interaction after

U_R rotation. Nonetheless, for U_L , since d_L and s_L have different quantum number, the rotation:

$$\begin{pmatrix} d_L \\ s_L \end{pmatrix} \rightarrow A_L \begin{pmatrix} d_L \\ s_L \end{pmatrix} \quad (342)$$

will move the mixing into the Charged Current (CC) and the Neutral Current (NC). One mixing angle θ remains physical (similar to the Cabibbo angle).

In summary, **we have 3 physical masses, and 1 rotation angle** describing the mixing between d_L and s_L in charged and neutral weak interactions. Since there is no phase angle, then we don't have CP violation.

Remark: Note that The U_L contain 3 phase angles and 1 rotation angle. However, we can use redefinition from u, d, s quarks to remove those phase angles, leaving the rotation angle. $\square$

Question 3

Jarlskog invariant and unitarity triangles

The Jarlskog invariant J is defined via

$$\text{Im}(V_{ij}V_{kl}V_{il}^*V_{kj}^*) = J \sum_{m,n=1}^3 \varepsilon_{ikm}\varepsilon_{jln}, \quad (343)$$

where $i, j, k, l = 1, 2, 3$ and ε_{ikm} is the totally antisymmetric Levi-Civita symbol.

- (a) After diagonalizing the quark mass matrices, the mass-eigenstate quark fields may still be redefined by independent phase rotations $d_{L,R}^j \rightarrow e^{i\alpha_j} d_{L,R}^j$ and $u_{L,R}^j \rightarrow e^{i\beta_j} u_{L,R}^j$, which leave the kinetic terms and masses unchanged. Determine how V_{ij} transforms under these rephasings, and show that $V_{ij}V_{kl}V_{il}^*V_{kj}^*$ is invariant. Explain why the imaginary part of such a rephasing-invariant quantity is a good candidate for a physical measure of CP violation.
- (b) Show, using the orthogonality of distinct rows and columns of the CKM matrix, that the area of all six unitarity triangles is exactly equal to $|J|/2$. *Note: (This holds for the unitarity triangles before normalizing one side to unit length.)*

Answer

(a)

First, in the charge weak current, the lagrangian is given by:

$$J_{CC}^\mu \propto \bar{u}_L^i V_{ij} \gamma^\mu d_L^j. \quad (344)$$

When we apply any phase rotation on the quark field, saying $d_{L,R}^j \rightarrow e^{i\alpha_j} d_{L,R}^j$ and $u_{L,R}^j \rightarrow e^{i\beta_j} u_{L,R}^j$, to keep the form of the charged-current interaction invariant, the CKM matrix V_{ij} transforms as

$$V_{ij} \rightarrow V'_{ij} = e^{i\beta_i} e^{-i\alpha_j} V_{ij}. \quad (345)$$

For the Jarlskog invariant J , after the rotation, it is given by:

$$V_{ij}V_{kl}V_{il}^*V_{kj}^* \rightarrow V'_{ij}V'_{kl}(V'_{il})^*(V'_{kj})^* \quad (346)$$

$$= (e^{i\beta_i} V_{ij} e^{-i\alpha_j}) (e^{i\beta_k} V_{kl} e^{-i\alpha_l}) (e^{-i\beta_i} V_{il}^* e^{i\alpha_l}) (e^{-i\beta_k} V_{kj}^* e^{i\alpha_j}) \quad (347)$$

$$= (e^{i\beta_i} e^{-i\beta_i}) (e^{i\beta_k} e^{-i\beta_k}) (e^{-i\alpha_j} e^{i\alpha_j}) (e^{-i\alpha_l} e^{i\alpha_l}) V_{ij}V_{kl}V_{il}^*V_{kj}^* \quad (348)$$

$$= V_{ij}V_{kl}V_{il}^*V_{kj}^*. \quad (349)$$

Hence, this J is invariant under the phase rotation. This property is helpful when we measure CP violation phenomenon, since it is independent of any phase redefinition for quark fields which absorb

unphysical phase. Under this property, the CP violation parameter cannot be removed by the phase redefinition.

(b)

Since the CKM matrix V is unitary, we can consider

$$V^\dagger V = \mathbf{1} \quad (350)$$

$$\implies V_{ki}V_{kj}^* = 0, \quad \text{if } i \neq j. \quad (351)$$

Without loss of generality, we can assume $i = b, j = d$, then we have

$$\underbrace{V_{ud}V_{ub}^*}_{z_1} + \underbrace{V_{cd}V_{cb}^*}_{z_2} + \underbrace{V_{td}V_{tb}^*}_{z_3} = 0 \quad (352)$$

$$\implies z_1 + z_2 + z_3 = 0. \quad (353)$$

Then the area A of this triangle is given by (compared definition $\text{Im}(V_{ij}V_{kl}V_{il}^*V_{kj}^*) = J \sum_{m,n=1}^3 \varepsilon_{ikm}\varepsilon_{jln}$)

$$A = \frac{1}{2} |\text{Im}(z_1 z_2^*)| \quad (354)$$

$$= \frac{1}{2} |\text{Im}(V_{ud}V_{ub}^*V_{cd}V_{cb}^*)| = \frac{1}{2} |\text{Im}(V_{ud}V_{cb}V_{ub}^*V_{cd}^*)| = \frac{1}{2} \left| J \sum_{m,n=1}^3 \varepsilon_{ucm}\varepsilon_{dbn} \right| \quad (355)$$

$$= \frac{1}{2} |J \varepsilon_{uct}\varepsilon_{dbs}| = \frac{1}{2} |J \varepsilon_{123}\varepsilon_{132}| = \frac{|J|}{2}. \quad (356)$$

Since we can choose any arbitrary indices i, j ($i \neq j$) for rows or columns, this result holds for all six unitarity triangles. Hence, we complete the proof that $A = \frac{|J|}{2}$. $\square$

Problem Set 6 Due 11am, Monday, April 20, 2026

Question 1

UV completion of the Weinberg operator without ν_R

Consider an extension of the Standard Model containing a heavy complex scalar triplet Δ transforming as $(1, 3, 1)$ under $SU(3)_C \times SU(2)_L \times U(1)_Y$. Write the triplet in matrix form as

$$\Delta = \begin{pmatrix} \Delta^+/\sqrt{2} & \Delta^{++} \\ \Delta^0 & -\Delta^+/\sqrt{2} \end{pmatrix}. \quad (357)$$

Let L_i denote the lepton doublets, where i, j are flavor indices, and define $\tilde{H} \equiv i\sigma_2 H^*$. Suppose the Lagrangian contains the renormalizable interactions

$$\mathcal{L} \supset -\frac{1}{2}\xi_{ij}\bar{L}_i^c i\sigma_2 \Delta L_j - \mu \tilde{H}^\dagger \Delta^\dagger H - M_\Delta^2 \text{Tr}(\Delta^\dagger \Delta) + h.c. \quad (358)$$

where ξ_{ij} are dimensionless couplings and μ, M_Δ are new mass parameters. This is known as the *type-II* seesaw mechanism.

- Show that the two interaction terms above are gauge invariant under $SU(2)_L \times U(1)_Y$. In particular, verify that the hypercharge assignments are consistent and explain why the flavor matrix ξ_{ij} must be symmetric. *Hint:* Recall that under an $SU(2)_L$ transformation U , $\Delta \rightarrow U\Delta U^\dagger$.
- Assume M_Δ is much larger than the electroweak scale. Integrate out the heavy triplet at tree level and show that the resulting low-energy effective theory contains the Weinberg operator

$$\mathcal{L}_{\text{eff}} \supset -\frac{\kappa_{ij}}{\Lambda}(\bar{L}_i \tilde{H})(\tilde{H}^T L_j^c) + h.c. \quad (359)$$

Determine the coefficient in terms of ξ_{ij}, μ and M_Δ , up to an overall sign and convention-dependent factors of 2.

- After electroweak symmetry breaking, determine the resulting neutrino mass matrix $(m_\nu)_{ij}$. Compare your result with the schematic right-handed neutrino seesaw scaling $m_\nu \approx \frac{y_\nu v^2}{M_N}$, or equivalently $\frac{\kappa}{\Lambda} \approx \frac{y_\nu^2}{M_N}$, which arises from integrating out heavy right-handed neutrinos with mass M_N and Yukawa coupling y_ν . What is the corresponding heavy mass scale and coupling combination in the scalar-triplet case?

Answer

(a)

For the second interaction term, $\tilde{H}^\dagger \Delta^\dagger H$, in order to have $U(1)_Y$ invariant, the hypercharge of Δ is

given by

$$\frac{1}{2} - Y_\Delta + \frac{1}{2} = 0 \quad (360)$$

$$\implies Y_\Delta = 1. \quad (361)$$

The corresponding quantum numbers or representation for $L, L^c, H, \tilde{H}, \Delta, \Delta^\dagger$ under the gauge group $SU(2)_L \times U(1)_Y$ are listed in Table 2.

Field	$SU(2)_L$	$U(1)_Y$
L	2	$-1/2$
L^c	2	$1/2$
H	2	$1/2$
$\tilde{H}$	2	$-1/2$
Δ	3	1
$\Delta^\dagger$	3	-1

Table 2: Field quantum numbers under $SU(2)_L \times U(1)_Y$.

Now, we can verify the invariance under each transformation:

- $U(1)_Y$:

$$-\frac{1}{2}\xi_{ij}\bar{L}_i^c i\sigma_2 \Delta L_j \quad (362)$$

$$\rightarrow -\frac{1}{2}\xi_{ij}(\bar{L}_i^c e^{-\alpha(x)/2})i\sigma_2(e^{\alpha(x)}\Delta)(e^{-\alpha(x)/2}L_j) \quad (363)$$

$$= -\frac{1}{2}\xi_{ij}\bar{L}_i^c i\sigma_2 \Delta L_j, \quad (364)$$

$$-\mu\tilde{H}^\dagger \Delta^\dagger H \quad (365)$$

$$\rightarrow -\mu(\tilde{H}^\dagger e^{\alpha(x)/2})(e^{-\alpha(x)}\Delta^\dagger)(e^{\alpha(x)/2}H) \quad (366)$$

$$= -\mu\tilde{H}^\dagger \Delta^\dagger H. \quad (367)$$

Hence, the interaction terms are invariant under $U(1)_Y$ gauge transformation.

- $SU(2)$: Note that $L^c = -i\gamma^2 L^* = -i\gamma^2 (L^\dagger)^T = -i\gamma^2 \gamma^0 \bar{L}^T = \mathbf{C}\bar{L}^T$, where γ^2 is the dirac matrix acting on the dirac spinor space, and $\mathbf{C}$ is the charge conjugation operator. Also, for $\tilde{H} = i\sigma_2 H^*$,

we have

$$\tilde{H} \rightarrow i\sigma_2(UH)^* \quad (368)$$

$$= i\sigma_2 U^* H^* = i\sigma_2 \left(e^{-i(\alpha^1 \tau_1 + \alpha^2 \tau_2 + \alpha^3 \tau_3)} \right)^* H^* \quad (369)$$

$$= \underbrace{i\sigma_2 e^{i(\alpha^1 \tau_1 - \alpha^2 \tau_2 + \alpha^3 \tau_3)}}_{e^{i(-\alpha^1 \tau_1 - \alpha^2 \tau_2 - \alpha^3 \tau_3)} i\sigma_2} H^* \quad (370)$$

$$= U i\sigma_2 H^* \quad (371)$$

$$= U \tilde{H} \quad (372)$$

Now we have,

$$L^c = -i\gamma_2 L^* \rightarrow -i\gamma_2 (UL)^* = -i\gamma_2 U^* L^* \quad (373)$$

$$= U^* (-i\gamma_2) L^* = U^* L^c \quad (374)$$

Now the first term under the $SU(2)_L$ transformation is given by

$$- \frac{1}{2} \xi_{ij} \bar{L}_i^c i\sigma_2 \Delta L_j \quad (375)$$

$$\rightarrow - \frac{1}{2} \xi_{ij} (\bar{L}_i^c (U^*)^\dagger) i\sigma_2 (U \Delta U^\dagger) U L_j \quad (376)$$

$$= - \frac{1}{2} \xi_{ij} \bar{L}_i^c U^T i\sigma_2 U \Delta L_j. \quad (377)$$

For $U^T i\sigma_2 U$, we can simplify it as

$$U^T i\sigma_2 U = (e^{-i\alpha^a \tau_a})^T i\sigma_2 e^{-i\alpha^a \tau_a} \quad (378)$$

$$= \underbrace{(e^{-i(\alpha^1 \tau_1 + \alpha^2 \tau_2 + \alpha^3 \tau_3)})^T}_{e^{-i(\alpha^1 \tau_1 - \alpha^2 \tau_2 + \alpha^3 \tau_3)}} \underbrace{i\sigma_2 e^{-i(\alpha^1 \tau_1 + \alpha^2 \tau_2 + \alpha^3 \tau_3)}}_{e^{-i(-\alpha^1 \tau_1 + \alpha^2 \tau_2 - \alpha^3 \tau_3)} i\sigma_2} \quad (379)$$

$$= i\sigma_2. \quad (380)$$

Hence, we have

$$- \frac{1}{2} \xi_{ij} \bar{L}_i^c i\sigma_2 \Delta L_j \quad (381)$$

$$\rightarrow - \frac{1}{2} \xi_{ij} \bar{L}_i^c U^T i\sigma_2 U \Delta L_j \quad (382)$$

$$= - \frac{1}{2} \xi_{ij} \bar{L}_i^c i\sigma_2 \Delta L_j. \quad (383)$$

For the second term, we have

$$- \mu \tilde{H}^\dagger \Delta^\dagger H \quad (384)$$

$$\rightarrow - \mu (\tilde{H}^\dagger U^\dagger) (U \Delta U^\dagger)^\dagger U H \quad (385)$$

$$= - \mu \tilde{H}^\dagger U^\dagger U \Delta^\dagger U^\dagger U H \quad (386)$$

$$= - \mu \tilde{H}^\dagger \Delta^\dagger H. \quad (387)$$

Hence, both interaction terms are invariant under $SU(2)_L$ and $U(1)_Y$ gauge transformations.

Next, for the discussion about ξ_{ij} , we first observe the term $i\sigma_2\Delta$, which is given by

$$M = i\sigma_2\Delta = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} \Delta^+/\sqrt{2} & \Delta^{++} \\ \Delta^0 & -\Delta^+/\sqrt{2} \end{pmatrix} = \begin{pmatrix} \Delta^0 & -\Delta^+/\sqrt{2} \\ -\Delta^+/\sqrt{2} & -\Delta^{++} \end{pmatrix} \quad (388)$$

$$\Rightarrow M = M^T. \quad (389)$$

Then if we take the transpose on $\bar{L}_i^c i\sigma_2\Delta L_j$, we can get

$$\bar{L}_i^c M L_j = (-i\gamma^2 L_i^*)^\dagger \gamma^0 M L_j = L_i^T (+i)(-\gamma^2) \gamma^0 M L_j = L_i^T (-i\gamma^2) \gamma^0 M L_j \quad (390)$$

$$\xRightarrow{\text{transpose}} \underbrace{(-1)}_{\text{since exchange two fermions}} (L_j)^T M^T \underbrace{(\gamma^0)^T}_{\gamma^0} \underbrace{(-i\gamma^2)^T}_{-i\gamma^2} L_i = L_j^T M^T \underbrace{\gamma^0 (i\gamma^2)}_{(-i\gamma^2)\gamma^0} L_i \quad (391)$$

$$= L_j^T \underbrace{M^T (-i\gamma^2) \gamma^0}_{[M, \gamma^\mu]=0} L_i \quad (392)$$

$$= L_j^T (-i\gamma^2) \gamma^0 M^T L_i = \bar{L}_j^c M L_i. \quad (393)$$

Hence, we have $\bar{L}_i^c M L_j = \bar{L}_j^c M L_i$, which is symmetric with interchanging i, j , and the ξ_{ij} has to be symmetric as well.

(b)

We first compute the amplitude for the process $L_j \bar{L}_i^c \rightarrow H \tilde{H}$ via s-channel scalar exchange, as shown in Fig. 4a. Here we ignore the spinor structure. The amplitude is given by

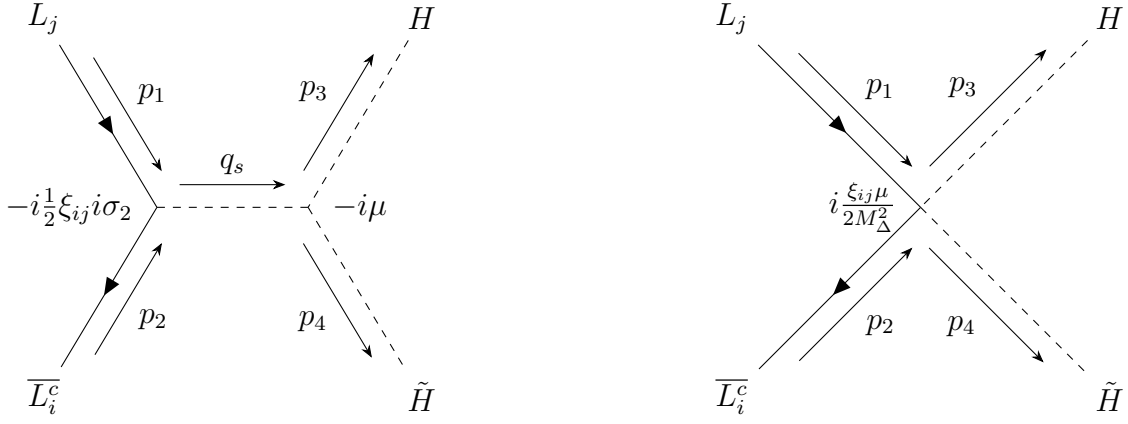
$$i\mathcal{M} = \left(-\frac{i}{2} \xi_{ij} \right) \frac{i}{q_s^2 - M_\Delta^2} (-i\mu) \quad (394)$$

$$= -i \frac{\xi_{ij} \mu}{2(q_s^2 - M_\Delta^2)}. \quad (395)$$

In the low-energy limit where $q_s^2 \ll M_\Delta^2$, we can approximate the amplitude as

$$i\mathcal{M} \approx i \frac{\xi_{ij} \mu}{2M_\Delta^2}. \quad (396)$$

This corresponds to an effective 4-point contact interaction, as shown in Fig. 4b, with an effective



(a) Feynman diagram for $L_j \bar{L}_i^c \rightarrow H \tilde{H}$ via s-channel scalar exchange. $q_s = p_1 + p_2 = p_3 + p_4$ is the momentum of the intermediate scalar. (b) Feynman diagram for $L_j \bar{L}_i^c \rightarrow H \tilde{H}$ via 4-point contact interaction. This diagram arises from integrating out the heavy scalar Δ in the limit $M_\Delta \rightarrow \infty$.

Figure 4: Tree-level Feynman diagrams for $L_j \bar{L}_i^c \rightarrow H \tilde{H}$ scattering via scalar interactions.

Lagrangian term given by

$$\mathcal{L}_{\text{eff}} \supset -\frac{\xi_{ij}\mu}{2M_\Delta^2} (\bar{L}_i \tilde{H}) (\tilde{H}^T L_j^c) + h.c. \quad (397)$$

Comparing this with the Weinberg operator, we can identify the coefficient as

$$\frac{\kappa_{ij}}{\Lambda} = \frac{\xi_{ij}\mu}{2M_\Delta^2}. \quad (398)$$

(c)

After electroweak symmetry breaking, the Higgs field acquires a vacuum expectation value $\langle H \rangle = (0, v/\sqrt{2})^T$. Substituting this into the effective Lagrangian term, we get the neutrino mass matrix as

$$-\frac{\xi_{ij}\mu}{2M_\Delta^2} (\bar{L}_i \tilde{H}) (\tilde{H}^T L_j^c) + h.c. \quad (399)$$

$$\Rightarrow -\frac{\xi_{ij}\mu}{2M_\Delta^2} \left(\bar{L}_i \begin{pmatrix} 0 \\ v/\sqrt{2} \end{pmatrix} \right) \left(\begin{pmatrix} 0 & v/\sqrt{2} \end{pmatrix} L_j^c \right) + h.c. \quad (400)$$

$$= -\frac{\xi_{ij}\mu v^2}{4M_\Delta^2} \bar{\nu}_{iL} \nu_{jL}^c + h.c. \quad (401)$$

$$= -\frac{\xi_{ij}\mu v^2}{4M_\Delta^2} \bar{\nu}_{iL} \nu_{jR} + h.c. \quad (402)$$

Hence, the neutrino mass matrix is given by

$$(m_\nu)_{ij} = \frac{\xi_{ij}\mu v^2}{4M_\Delta^2}. \quad (403)$$

Comparing this with the right-handed neutrino seesaw scaling $m_\nu \approx \frac{y_\nu v^2}{M_N}$, we can identify the corre-

sponding heavy mass scale and coupling combination in the scalar-triplet case as

$$M_N \leftrightarrow \frac{4M_\Delta^2}{\xi_{ij}\mu}, \quad y_\nu \leftrightarrow 1. \quad (404)$$

□

Question 2

Majorana phases and neutrino oscillations

The PMNS matrix for Majorana neutrinos is $U_{\text{PMNS}} = U_{\text{Dirac}} \text{diag}(1, e^{i\alpha/2}, e^{i\beta/2})$, where U_{Dirac} contains three mixing angles and one Dirac CP-violating phase, while α and β are Majorana phases.

- Write the vacuum oscillation amplitude for $\nu_l \rightarrow \nu_{l'}$ in terms of U_{PMNS} , the neutrino masses, the baseline L , and the neutrino energy E . Show explicitly that the Majorana phases cancel from the oscillation probability.
- Briefly explain why Majorana phases can enter lepton-number-violating processes such as neutrinoless double beta decay, but do not affect ordinary neutrino oscillations.

Answer

(a)

The flavor eigenstates of neutrinos can be expressed as a linear combination of the mass eigenstates as follows:

$$|\nu_l\rangle = \sum_{j=1}^3 U_{lj}^* |\nu_j\rangle, \quad (405)$$

where U_{lj} are the elements of the PMNS matrix. The time evolution of the mass eigenstates is given by

$$|\nu_j(t)\rangle = e^{-iE_j t} |\nu_j\rangle, \quad (406)$$

where E_j is the energy of the j -th mass eigenstate. The flavor state at time t can be written as

$$|\nu_l(t)\rangle = \sum_{j=1}^3 U_{lj}^* e^{-iE_j t} |\nu_j\rangle. \quad (407)$$

The oscillation amplitude for $\nu_l \rightarrow \nu_{l'}$ can be calculated as follows:

$$A_{l \rightarrow l'}(t) = \langle \nu_{l'} | \nu_l(t) \rangle \quad (408)$$

$$= \sum_{i=1}^3 U_{l'i} \langle \nu_i | \sum_{j=1}^3 U_{lj}^* e^{-iE_j t} |\nu_j\rangle \quad (409)$$

$$= \sum_{i=1}^3 U_{l'i} U_{li}^* e^{-iE_i t}. \quad (410)$$

Now, we can express the PMNS matrix in terms of the Dirac and Majorana phases as follows:

$$U_{\text{PMNS}} = U_{\text{Dirac}} \cdot \text{diag}(1, e^{i\alpha/2}, e^{i\beta/2}). \quad (411)$$

This means that the elements of the PMNS matrix can be written as

$$U_{li} = (U_{\text{Dirac}})_{li} \cdot e^{i\phi_i}, \quad (412)$$

where ϕ_i is the Majorana phase associated with the i -th mass eigenstate. Substituting this into the expression for the oscillation amplitude, we get

$$A_{l \rightarrow l'}(t) = \sum_{i=1}^3 (U_{\text{Dirac}})_{l'i} e^{i\phi_i} (U_{\text{Dirac}})_{li}^* e^{-i\phi_i} e^{-iE_i t} \quad (413)$$

$$= \sum_{i=1}^3 (U_{\text{Dirac}})_{l'i} (U_{\text{Dirac}})_{li}^* e^{-iE_i t}. \quad (414)$$

As we can see, the Majorana phases $e^{i\phi_i}$ and $e^{-i\phi_i}$ cancel out in the expression for the oscillation amplitude. Therefore, the oscillation probability, which is given by $P_{l \rightarrow l'}(t) = |A_{l \rightarrow l'}(t)|^2$, is also independent of the Majorana phases.

(b)

Since Majorana phases do not affect the oscillation probabilities, they cannot be probed through ordinary neutrino oscillation experiments. If neutrinos are Dirac particles, then the process should conserve lepton number. However, if neutrinos are Majorana particles, then lepton number is not conserved, and processes that violate lepton number is possible. Hence, the Majorana phases can enter lepton-number-violating processes such as neutrinoless double beta decay.

Question 3

Solar neutrino day-night effect

Solar neutrino oscillations can be modified by the fact that during the night neutrinos must pass through the Earth before reaching the detectors. Hence the oscillation probability can differ between neutrinos arriving during the day and at night. Experiments with real-time event reconstruction can therefore search for a day-night asymmetry in the measured solar neutrino flux.

To understand this effect, assume that high-energy ${}^8\text{B}$ solar neutrinos arrive at the surface of the Earth approximately in the $|\nu_2\rangle$ mass eigenstate. This is a consequence of adiabatic MSW conversion in the Sun, and for the physical value $\Delta m_{21}^2 \approx 7.5 \times 10^{-5} \text{ eV}^2$ the approximation is still quite good, at the few-percent level. Work in a two-flavor framework (ν_e, ν_x) where $|\nu_x\rangle$ is the active flavor state orthogonal to $|\nu_e\rangle$.

- By expressing $|\nu_2\rangle$ as a linear combination of $|\nu_e\rangle$ and $|\nu_x\rangle$, where the mixing angle is defined by $|\nu_e\rangle = \cos\theta|\nu_1\rangle + \sin\theta|\nu_2\rangle$, compute the probability that $|\nu_2\rangle$ is detected as an electron neutrino.
- If instead the solar neutrino, originally in the $|\nu_2\rangle$ state, propagates a distance L through the Earth before reaching the detector, compute the probability that this neutrino is detected as an electron neutrino, assuming that the electron number density n_e is constant along the neutrino path.

Hint: In matter of constant density, show that the oscillation frequency is

$$\Delta_M = \sqrt{\Delta^2 \sin^2 2\theta + (\Delta \cos 2\theta - \sqrt{2}G_F n_e)^2}, \quad (415)$$

where $\Delta = (\Delta m^2)/(2E_\nu)$, and that the matter mixing angle satisfies $\Delta_M \sin 2\theta_M = \Delta \sin 2\theta$ and $\Delta_M \cos 2\theta_M = \Delta \cos 2\theta - \sqrt{2}G_F n_e$.

- Given the values $L = 3000 \text{ km}$, $\sqrt{2}G_F n_e = 1.5 \times 10^{-13} \text{ eV}$, $\Delta m_{21}^2 = 7.5 \times 10^{-5} \text{ eV}^2$, $\sin^2 \theta \approx 0.3$ and $E_\nu = 8 \text{ MeV}$, obtain $P_{ee}^{\text{night}} - P_{ee}^{\text{day}}$. Compare your result qualitatively with the observed fact that the solar-neutrino day-night asymmetry is at the few-percent level. In 2014, the Super-Kamiokande experiment reported a 2.7σ indication of a nonzero day-night asymmetry for ${}^8\text{B}$ neutrinos, which has been updated to a 2.9σ effect in Phys. Rev. **D94**, 052010 (2016).

Answer

(a)

Given that $|\nu_e\rangle = \cos\theta|\nu_1\rangle + \sin\theta|\nu_2\rangle$, we can express $|\nu_2\rangle$ as a linear combination of $|\nu_e\rangle$ and $|\nu_x\rangle$ as follows:

$$|\nu_2\rangle = \sin\theta|\nu_e\rangle + \cos\theta|\nu_x\rangle. \quad (416)$$

The probability that $|\nu_2\rangle$ is detected as an electron neutrino is given by the square of the amplitude for $|\nu_2\rangle$ to be in the $|\nu_e\rangle$ state:

$$P_{ee}^{\text{day}} = |\langle \nu_e | \nu_2 \rangle|^2 = |\sin \theta|^2 = \sin^2 \theta. \quad (417)$$

(b)

To derive Δ_M and θ_M , we start with the effective Hamiltonian for neutrino propagation in matter in the flavor basis ($|\nu_e\rangle, |\nu_x\rangle$):

$$H_f = H_{vac} + H_{int} \quad (418)$$

The vacuum part H_{vac} , after subtracting a common trace term $\frac{E_1+E_2}{2}$, is given by:

$$H_{vac} = \frac{\Delta}{2} \begin{pmatrix} -\cos 2\theta & \sin 2\theta \\ \sin 2\theta & \cos 2\theta \end{pmatrix}, \quad (419)$$

where $\Delta = \frac{\Delta m^2}{2E}$. The interaction part H_{int} arises from the coherent forward scattering of ν_e on electrons (via W exchange), contributing a potential $V = \sqrt{2}G_F n_e$:

$$H_{int} = \begin{pmatrix} V & 0 \\ 0 & 0 \end{pmatrix}. \quad (420)$$

Total Hamiltonian H_f (after again subtracting $V/2$ to make it traceless) becomes:

$$H_f = \frac{1}{2} \begin{pmatrix} V - \Delta \cos 2\theta & \Delta \sin 2\theta \\ \Delta \sin 2\theta & -(V - \Delta \cos 2\theta) \end{pmatrix}. \quad (421)$$

We wish to express this in the same form as the vacuum Hamiltonian but with matter-modified parameters Δ_M and θ_M :

$$H_f = \frac{\Delta_M}{2} \begin{pmatrix} -\cos 2\theta_M & \sin 2\theta_M \\ \sin 2\theta_M & \cos 2\theta_M \end{pmatrix}. \quad (422)$$

Comparing the matrix elements, we obtain the following relations:

1. Off-diagonal term:

$$\Delta_M \sin 2\theta_M = \Delta \sin 2\theta \quad (423)$$

2. Diagonal term:

$$-\Delta_M \cos 2\theta_M = V - \Delta \cos 2\theta \implies \Delta_M \cos 2\theta_M = \Delta \cos 2\theta - V \quad (424)$$

To find the oscillation frequency Δ_M , we square and add these two equations:

$$\begin{aligned}\Delta_M^2(\sin^2 2\theta_M + \cos^2 2\theta_M) &= (\Delta \sin 2\theta)^2 + (\Delta \cos 2\theta - V)^2 \\ \Delta_M &= \sqrt{\Delta^2 \sin^2 2\theta + (\Delta \cos 2\theta - \sqrt{2}G_F n_e)^2}.\end{aligned}\quad (425)$$

The matter mixing angle θ_M is then implicitly defined by:

$$\tan 2\theta_M = \frac{\Delta \sin 2\theta}{\Delta \cos 2\theta - \sqrt{2}G_F n_e}.\quad (426)$$

In the Earth's matter with constant electron density n_e , the effective Hamiltonian eigenstates are the matter eigenstates $|\nu_{1m}\rangle$ and $|\nu_{2m}\rangle$. These are related to the flavor states by the matter mixing angle θ_M :

$$|\nu_e\rangle = \cos \theta_M |\nu_{1m}\rangle + \sin \theta_M |\nu_{2m}\rangle,\quad (427)$$

$$|\nu_x\rangle = -\sin \theta_M |\nu_{1m}\rangle + \cos \theta_M |\nu_{2m}\rangle.\quad (428)$$

The solar neutrino enters the Earth in the $|\nu_2\rangle$ mass eigenstate. Expressing $|\nu_2\rangle$ in terms of the matter eigenstates:

$$\begin{aligned}|\nu(0)\rangle &= |\nu_2\rangle = \sin \theta |\nu_e\rangle + \cos \theta |\nu_x\rangle \\ &= \sin \theta (\cos \theta_M |\nu_{1m}\rangle + \sin \theta_M |\nu_{2m}\rangle) + \cos \theta (-\sin \theta_M |\nu_{1m}\rangle + \cos \theta_M |\nu_{2m}\rangle) \\ &= -\sin(\theta_M - \theta) |\nu_{1m}\rangle + \cos(\theta_M - \theta) |\nu_{2m}\rangle.\end{aligned}\quad (429)$$

After propagating a distance L , the state evolves to (ignoring a global phase):

$$|\nu(L)\rangle = -e^{-iE_{1m}L} \sin(\theta_M - \theta) |\nu_{1m}\rangle + e^{-iE_{2m}L} \cos(\theta_M - \theta) |\nu_{2m}\rangle,\quad (430)$$

where $E_{2m} - E_{1m} = \Delta_M$ is the oscillation frequency in matter. The probability of detecting this neutrino as an electron neutrino is $P_{ee}^{\text{night}} = |\langle \nu_e | \nu(L) \rangle|^2$:

$$\begin{aligned}\langle \nu_e | \nu(L) \rangle &= -\cos \theta_M \sin(\theta_M - \theta) + e^{-i\Delta_M L} \sin \theta_M \cos(\theta_M - \theta) \\ &= -\cos \theta_M (\sin \theta_M \cos \theta - \cos \theta_M \sin \theta) + e^{-i\Delta_M L} \sin \theta_M (\cos \theta_M \cos \theta + \sin \theta_M \sin \theta).\end{aligned}\quad (431)$$

Using the identity $|\alpha + \beta e^{-i\phi}|^2 = \alpha^2 + \beta^2 + 2\alpha\beta \cos \phi$, and simplifying the result:

$$P_{ee}^{\text{night}} = \sin^2 \theta + \sin 2\theta_M \sin 2(\theta_M - \theta) \sin^2 \left(\frac{\Delta_M L}{2} \right).\quad (432)$$

The second term represents the regeneration of the electron neutrino flavor due to the Earth's matter effect.

(c)

First, we calculate the vacuum oscillation parameter Δ :

$$\Delta = \frac{\Delta m_{21}^2}{2E_\nu} = \frac{7.5 \times 10^{-5} \text{ eV}^2}{2 \times 8 \times 10^6 \text{ eV}} \approx 4.69 \times 10^{-12} \text{ eV}. \quad (433)$$

The matter potential is given by $V = \sqrt{2}G_F n_e = 1.5 \times 10^{-13} \text{ eV}$. Given $\sin^2 \theta \approx 0.3$, we have $\cos 2\theta = 1 - 2\sin^2 \theta = 0.4$ and $\sin 2\theta = \sqrt{1 - 0.4^2} \approx 0.917$. The matter oscillation parameters are:

$$\Delta_M \sin 2\theta_M = \Delta \sin 2\theta \approx 4.30 \times 10^{-12} \text{ eV}, \quad (434)$$

$$\Delta_M \cos 2\theta_M = \Delta \cos 2\theta - V \approx 1.88 \times 10^{-12} - 0.15 \times 10^{-12} = 1.73 \times 10^{-12} \text{ eV}. \quad (435)$$

The oscillation frequency in matter is $\Delta_M = \sqrt{(4.30)^2 + (1.73)^2} \times 10^{-12} \approx 4.63 \times 10^{-12} \text{ eV}$. The mixing angles satisfy:

$$\sin 2\theta_M \approx 0.928, \quad \cos 2\theta_M \approx 0.373. \quad (436)$$

The factor for the asymmetry is:

$$\sin 2(\theta_M - \theta) = \sin 2\theta_M \cos 2\theta - \cos 2\theta_M \sin 2\theta \approx (0.928)(0.4) - (0.373)(0.917) \approx 0.029. \quad (437)$$

For solar neutrinos, the oscillation phase is averaged due to the energy spread, so $\langle \sin^2(\Delta_M L/2) \rangle = 1/2$.

The day-night difference is:

$$\begin{aligned} P_{ee}^{\text{night}} - P_{ee}^{\text{day}} &= \frac{1}{2} \sin 2\theta_M \sin 2(\theta_M - \theta) \\ &\approx \frac{1}{2} (0.928)(0.029) \approx 0.0135 = 1.35\%. \end{aligned} \quad (438)$$

□

Problem Set 7 Due 11am, Monday, May 4, 2026

Question 1

Comparing Z-boson partial widths

For massless fermions, the tree-level decay width of the Z boson is

$$\Gamma(Z \rightarrow f\bar{f}) = N_f^c \frac{G_F M_Z^3}{6\sqrt{2}\pi} \left((g_V^f)^2 + (g_A^f)^2 \right), \quad (439)$$

where $N_f^c = 1$ for leptons and $N_f^c = 3$ for quarks, and

$$g_V^f = T_3^f - 2Q_f \sin^2 \theta_W, \quad g_A^f = T_3^f. \quad (440)$$

Take $\sin^2 \theta_W = 0.231$, and neglect fermion masses and radiative corrections.

- (a) For $f = \nu, e, u, d, b$, list T_3^f , and Q_f , and compute g_V^f , g_A^f , and $(g_V^f)^2 + (g_A^f)^2$.
- (b) Rank the partial widths

$$\Gamma(Z \rightarrow \nu\bar{\nu}), \quad \Gamma(Z \rightarrow e^+e^-), \quad \Gamma(Z \rightarrow u\bar{u}), \quad \Gamma(Z \rightarrow d\bar{d}), \quad \Gamma(Z \rightarrow b\bar{b}) \quad (441)$$

from largest to smallest, indicating any equalities.

- (c) Briefly explain your ranking. In particular, explain why $\Gamma(Z \rightarrow e^+e^-)$ is smaller than $\Gamma(Z \rightarrow \nu\bar{\nu})$, and why quark final states make the dominant contribution to the total Z width.

Answer

(a)

The values of T_3^f , Q_f , g_V^f , g_A^f , and $(g_V^f)^2 + (g_A^f)^2$ for $f = \nu, e, u, d, b$ are listed in Table 3.

f	T_3^f	Q_f	$g_V^f = T_3^f - 2Q_f \sin^2 \theta_W$	$g_A^f = T_3^f$	$(g_V^f)^2 + (g_A^f)^2$
ν	+1/2	0	+1/2	+1/2	0.5000
e	-1/2	-1	-1/2 + 2 sin ² θ_W = -0.038	-1/2	0.2514
u	+1/2	2/3	+1/2 - 4/3 sin ² θ_W = 0.192	+1/2	0.2868
d	-1/2	-1/3	-1/2 + 2/3 sin ² θ_W = -0.346	-1/2	0.3697
b	-1/2	-1/3	-1/2 + 2/3 sin ² θ_W = -0.346	-1/2	0.3697

Table 3: Values of T_3^f , Q_f , g_V^f , g_A^f , and $(g_V^f)^2 + (g_A^f)^2$ for $f = \nu, e, u, d, b$.

(b)

Since the partial width is proportional to $N_f^c((g_V^f)^2 + (g_A^f)^2)$, we have

$$\Gamma(Z \rightarrow \nu\bar{\nu}) \propto 1 \times 0.5000 = 0.5000, \quad (442)$$

$$\Gamma(Z \rightarrow e^+e^-) \propto 1 \times 0.2514 = 0.2514, \quad (443)$$

$$\Gamma(Z \rightarrow u\bar{u}) \propto 3 \times 0.2868 = 0.8604, \quad (444)$$

$$\Gamma(Z \rightarrow d\bar{d}) \propto 3 \times 0.3697 = 1.1091, \quad (445)$$

$$\Gamma(Z \rightarrow b\bar{b}) \propto 3 \times 0.3697 = 1.1091. \quad (446)$$

Thus, the ranking from largest to smallest is

$$\Gamma(Z \rightarrow d\bar{d}) \approx \Gamma(Z \rightarrow b\bar{b}) > \Gamma(Z \rightarrow u\bar{u}) > \Gamma(Z \rightarrow \nu\bar{\nu}) > \Gamma(Z \rightarrow e^+e^-). \quad (447)$$

Note: Of course, we can include other quark flavors (s, c) in the ranking as well. Then the ranking from largest to smallest is

$$\Gamma(Z \rightarrow d\bar{d}) \approx \Gamma(Z \rightarrow s\bar{s}) \approx \Gamma(Z \rightarrow b\bar{b}) > \Gamma(Z \rightarrow u\bar{u}) \approx \Gamma(Z \rightarrow c\bar{c}) > \Gamma(Z \rightarrow \nu\bar{\nu}) > \Gamma(Z \rightarrow e^+e^-). \quad (448)$$

Here, we ignore the mass differences between the quark flavors, which is a good approximation for u, d, s quarks, but not a good approximation for c, b quarks. Also, we ignore the loop-level corrections to the Z .

(c)

The partial width $\Gamma(Z \rightarrow e^+e^-)$ is smaller than $\Gamma(Z \rightarrow \nu\bar{\nu})$ because the vector coupling g_V^e is suppressed by the factor $1 - 4\sin^2\theta_W \approx 0.076$, which is close to zero, while g_V^ν is not suppressed.

Besides, because of the color factor $N_f^c = 3$ for quarks, which enhances the partial widths for quark final states compared to leptonic final states, quark final states make the dominant contribution to the total Z width. Including all possible quark flavors (u, d, s, c, b) further increases the total width from quark final states, making it even more dominant compared to leptonic final states. $\square$

Question 2

Higgs potential with a higher-dimension correction

The Standard Model Higgs potential can be modified by higher-dimension operators that parameterize possible effects of heavy new physics. As a simple example, consider

$$V(H) = -\mu^2 H^\dagger H + \lambda (H^\dagger H)^2 + \frac{c_6}{\Lambda^2} (H^\dagger H)^3, \quad (449)$$

where $\mu^2 > 0$, $\lambda > 0$, and c_6 is a dimensionless coupling, and Λ is a heavy mass scale. Assume this is the only relevant higher-dimension correction to the Higgs potential. Work in unitary gauge

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ v + h \end{pmatrix}. \quad (450)$$

- (a) Derive the condition that determines the vacuum expectation value v by minimizing the potential.
- (b) Expand the potential to quadratic and cubic order in h , and derive the physical Higgs mass m_h and the trilinear Higgs self-coupling λ_3 , defined by $V(h) \supset \frac{\lambda_3}{3!} h^3$.
- (c) Explain why measurements of v and m_h alone do not uniquely determine the coefficient c_6/Λ^2 . Define

$$\kappa_\lambda \equiv \frac{\lambda_3}{\lambda_3^{\text{SM}}}. \quad (451)$$

The High-Luminosity LHC (HL-LHC) is projected to measure the Higgs trilinear self-coupling with better than 30% precision. Assuming $c_6 \approx 1$ and taking $|\kappa_\lambda - 1| < 0.3$ estimate the corresponding lower bound on the heavy scale Λ . Use $v = 246$ GeV and $m_h = 125$ GeV.

Answer

(a)

To determine the vacuum expectation value v , we minimize the potential $V(H)$ with respect to H . In unitary gauge, we have

$$V(h) = -\frac{\mu^2}{2}(v+h)^2 + \frac{\lambda}{4}(v+h)^4 + \frac{c_6}{8\Lambda^2}(v+h)^6. \quad (452)$$

The condition for the minimum is given by

$$\left. \frac{dV}{dh} \right|_{h=0} = 0 \quad (453)$$

$$\left. \frac{d^2V}{dh^2} \right|_{h=0} > 0. \quad (454)$$

Both conditions are given by

$$-\mu^2(v+h) + \lambda(v+h)^3 + \frac{3}{4} \frac{c_6}{\Lambda^2} (v+h)^5 \Big|_{h=0} = 0 \quad (455)$$

$$-\mu^2 + 3\lambda(v+h)^2 + \frac{15}{4} \frac{c_6}{\Lambda^2} (v+h)^4 \Big|_{h=0} > 0. \quad (456)$$

Then, after setting $h = 0$, we have

$$-\mu^2 v + \lambda v^3 + \frac{3}{4} \frac{c_6}{\Lambda^2} v^5 = 0 = -\mu^2 + \lambda v^2 + \frac{3}{4} \frac{c_6}{\Lambda^2} v^4, \quad (457)$$

$$-\mu^2 + 3\lambda v^2 + \frac{15}{4} \frac{c_6}{\Lambda^2} v^4 > 0. \quad (458)$$

After rearranging the first equation, we get

$$0 = \frac{3}{4} \frac{c_6}{\Lambda^2} v^4 + \lambda v^2 - \mu^2 \quad (459)$$

$$\Rightarrow v^2 = \frac{-\lambda \pm \sqrt{\lambda^2 + 3c_6\mu^2/\Lambda^2}}{3c_6/(2\Lambda^2)} = \frac{2\Lambda^2}{3c_6} \left(-\lambda \pm \sqrt{\lambda^2 + \frac{3c_6\mu^2}{\Lambda^2}} \right) \quad (460)$$

$$= \frac{2\Lambda^2}{3c_6} \left(-\lambda \pm \lambda \sqrt{1 + \frac{3c_6\mu^2}{\lambda^2\Lambda^2}} \right) \quad (461)$$

$$= \frac{2\Lambda^2}{3c_6} \lambda \left(\pm \sqrt{1 + \frac{3c_6\mu^2}{\lambda^2\Lambda^2}} - 1 \right). \quad (462)$$

This solution is valid for $c_6 > 0$. For $c_6 < 0$, the solution is valid only if $1 + \frac{3c_6\mu^2}{\lambda^2\Lambda^2} > 0$, which requires $c_6 > -\lambda^2\Lambda^2/(3\mu^2)$. Also, this result should go back to the SM result $v^2 = \mu^2/\lambda$ in the limit $c_6/\Lambda^2 \rightarrow 0$. This requires us to take the solution with the plus sign, which gives

$$v^2 = \frac{2\Lambda^2}{3c_6} \lambda \left(\sqrt{1 + \frac{3c_6\mu^2}{\lambda^2\Lambda^2}} - 1 \right). \quad (463)$$

Now, we can take a look at the second condition. We can rearrange the second condition as

$$-\mu^2 + 3\lambda v^2 + \frac{15}{4} \frac{c_6}{\Lambda^2} v^4 \quad (464)$$

$$= \underbrace{\left(-\mu^2 + \lambda v^2 + \frac{3}{4} \frac{c_6}{\Lambda^2} v^4\right)}_{=0} + 2\lambda v^2 + 3 \frac{c_6}{\Lambda^2} v^4 \quad (465)$$

$$= v^2 \left(2\lambda + 3 \frac{c_6}{\Lambda^2} v^2\right) = v^2 \lambda \left(2 + \frac{3c_6}{\lambda \Lambda^2} v^2\right) \quad (466)$$

$$= v^2 \lambda \left(2 + \frac{3c_6}{\lambda \Lambda^2} \frac{2\Lambda^2}{3c_6} \lambda \left(\sqrt{1 + \frac{3c_6 \mu^2}{\lambda^2 \Lambda^2}} - 1\right)\right) \quad (467)$$

$$= v^2 \lambda \left(2 + 2 \left(\sqrt{1 + \frac{3c_6 \mu^2}{\lambda^2 \Lambda^2}} - 1\right)\right) \quad (468)$$

$$= 2v^2 \lambda \sqrt{1 + \frac{3c_6 \mu^2}{\lambda^2 \Lambda^2}} > 0. \quad (469)$$

Hence, the second condition is automatically satisfied for any $c_6 > -\lambda^2 \Lambda^2 / (3\mu^2)$.

(b)

To derive the physical Higgs mass m_h and the trilinear Higgs self-coupling λ_3 , we can take the second and third derivatives of the potential $V(h)$ with respect to h at $h = 0$, which gives

$$m_h^2 = \left. \frac{d^2 V}{dh^2} \right|_{h=0} = -\mu^2 + 3\lambda(v+h)^2 + \frac{15}{4} \frac{c_6}{\Lambda^2} (v+h)^4 \Big|_{h=0} = -\mu^2 + 3\lambda v^2 + \frac{15}{4} \frac{c_6}{\Lambda^2} v^4, \quad (470)$$

$$= 2\lambda v^2 + 3 \frac{c_6}{\Lambda^2} v^4, \quad (471)$$

$$v\lambda_3 = \left. \frac{d^3 V}{dh^3} \right|_{h=0} = 6\lambda(v+h) + 15 \frac{c_6}{\Lambda^2} (v+h)^3 \Big|_{h=0} = 6\lambda v + 15 \frac{c_6}{\Lambda^2} v^3. \quad (472)$$

Hence, we have

$$m_h^2 = 2\lambda v^2 + 3 \frac{c_6}{\Lambda^2} v^4, \quad (473)$$

$$\lambda_3 = 6\lambda + 15 \frac{c_6}{\Lambda^2} v^2. \quad (474)$$

with the corresponding lagrangian term:

$$V(h) \supset \frac{\lambda_3}{3!} v h^3 = \frac{1}{3!} \left(6\lambda v + 15 \frac{c_6}{\Lambda^2} v^3\right) h^3. \quad (475)$$

$$V(h) \supset \frac{1}{2} m_h^2 h^2 = \frac{1}{2} \left(2\lambda v^2 + 3 \frac{c_6}{\Lambda^2} v^4\right) h^2. \quad (476)$$

(c)

Measurements of v and m_h alone do not uniquely determine the coefficient c_6/Λ^2 because both m_h^2 and

λ_3 depend on the combination of λ and c_6/Λ^2 . Specifically, we have two equations with three unknowns (λ , c_6/Λ^2 , and Λ), which means we cannot solve for c_6/Λ^2 uniquely without additional information.

Before we estimate the coupling ratio κ_λ , we can express λ_3 and λ_3^{SM} in terms of m_h^2 , v^2 , and c_6/Λ^2 using the expression for m_h^2 to eliminate λ . We have

$$\lambda_3 = 6\lambda + 15\frac{c_6}{\Lambda^2}v^2 = \frac{3}{v^2} \left(\underbrace{2\lambda v^2 + 3\frac{c_6}{\Lambda^2}v^4}_{m_h^2} + 2\frac{c_6}{\Lambda^2}v^4 \right) \quad (477)$$

$$= \frac{3}{v^2} \left(m_h^2 + 2\frac{c_6}{\Lambda^2}v^4 \right), \quad (478)$$

$$\lambda_3^{\text{SM}} = 6\lambda = \frac{3}{v^2} (2\lambda v^2) = \frac{3}{v^2} m_h^2. \quad (479)$$

Then, the coupling ratio κ_λ is given by

$$\kappa_\lambda = \frac{\lambda_3}{\lambda_3^{\text{SM}}} = \frac{m_h^2 + 2\frac{c_6}{\Lambda^2}v^4}{m_h^2} = 1 + 2\frac{c_6}{\Lambda^2} \frac{v^4}{m_h^2}. \quad (480)$$

Assuming $c_6 \approx 1$ and taking $|\kappa_\lambda - 1| < 0.3$, we can estimate the corresponding lower bound on the heavy scale Λ as

$$0.3 > |\kappa_\lambda - 1| = 2\frac{v^4}{m_h^2} \frac{1}{\Lambda^2} \quad (481)$$

$$\Rightarrow \Lambda > \frac{v^2}{m_h} \sqrt{\frac{2}{0.3}} \approx 1.250 \text{ TeV}. \quad (482)$$

□

Question 3

Grand unification and proton decay

Grand unified theories (GUTs) unify the Standard Model gauge groups into a single simple gauge group at a high scale M_{GUT} . In a one-loop approximation, the gauge couplings evolve as

$$\alpha_i^{-1}(\mu) = \alpha_i^{-1}(M_Z) - \frac{b_i}{2\pi} \ln \left(\frac{\mu}{M_Z} \right), \quad (483)$$

where $i = 1, 2, 3$ correspond to the $U(1)_Y$, $SU(2)_L$, and $SU(3)_C$ gauge groups, respectively, and $\alpha_i = g_i^2/(4\pi)$ with $\alpha_1 = (5/3)\alpha_Y$. Assume $\alpha_1^{-1}(M_Z) = 59$, $\alpha_2^{-1}(M_Z) = 29.6$, and $\alpha_3^{-1}(M_Z) = 8.5$. The one-loop Standard Model coefficients are $b_1 = 41/10$, $b_2 = -19/6$, and $b_3 = -7$.

- (a) Sketch $\alpha_i^{-1}(\mu)$ as a function of $\ln(\mu)$ from M_Z up to $\mu \approx 10^{17}$ GeV. Do the three couplings unify at a single scale in the Standard Model? Estimate from your sketch the scale at which α_1 and α_2 meet.
- (b) In $SU(5)$, the heavy X, Y gauge bosons couple quarks to leptons. For example, one has schematic couplings $g_{\text{GUT}} X_\mu Q^\dagger \bar{\sigma}^\mu e_R^c$ and $g_{\text{GUT}} X_\mu u_R^c \bar{\sigma}^\mu Q$. Show that these are consistent with the Standard Model quantum numbers of the heavy gauge boson, and that integrating out a heavy X boson of mass M_X generates a low-energy dimension-6 operator of the form

$$\mathcal{L}_{\text{eff}} \supset \frac{g_{\text{GUT}}^2}{M_X^2} (QQ)(e_R^c u_R^c), \quad (484)$$

up to contractions. Show that this operator is baryon-number violating and can mediate proton decay.

- (c) Using dimensional analysis, show that the proton decay rate induced by this operator scales as $\Gamma_p \sim \alpha_{\text{GUT}}^2 M_p^5 / M_X^4$, where $\alpha_{\text{GUT}} \equiv g_{\text{GUT}}^2/(4\pi)$. Hence estimate the proton lifetime for $M_X \approx 10^{15}$ GeV, taking $M_p = 1$ GeV and $\alpha_{\text{GUT}} \approx 1/25$. Express your answer in years. Compare your result with the current lower limit $\tau(p \rightarrow e^+ \pi^0) \geq 10^{34}$ years. Finally, use your scaling result to explain why the lower bound on M_X depends only weakly on the experimental lower bound on the proton lifetime.

Answer

(a)

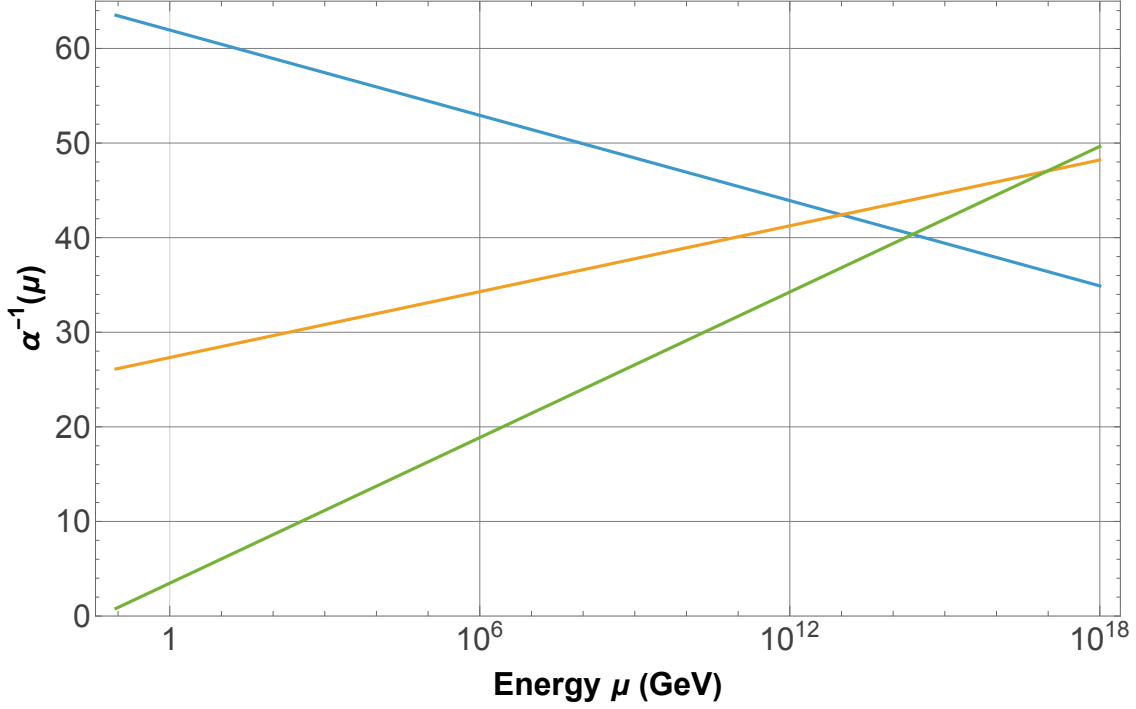


Figure 5: Sketch of $\alpha_i^{-1}(\mu)$ as a function of $\ln(\mu)$ from M_Z up to $\mu \approx 10^{17}$ GeV.

First, we calculate their numerical values:

$$\alpha_1^{-1}(\mu) = 59 - \frac{41}{20\pi} \ln\left(\frac{\mu}{M_Z}\right) \quad (485)$$

$$= -\frac{41}{20\pi} \ln(\mu) + \frac{41}{20\pi} \ln(M_Z) + 59, \quad (486)$$

$$\alpha_2^{-1}(\mu) = 29.6 + \frac{19}{12\pi} \ln\left(\frac{\mu}{M_Z}\right) \quad (487)$$

$$= \frac{19}{12\pi} \ln(\mu) - \frac{19}{12\pi} \ln(M_Z) + 29.6, \quad (488)$$

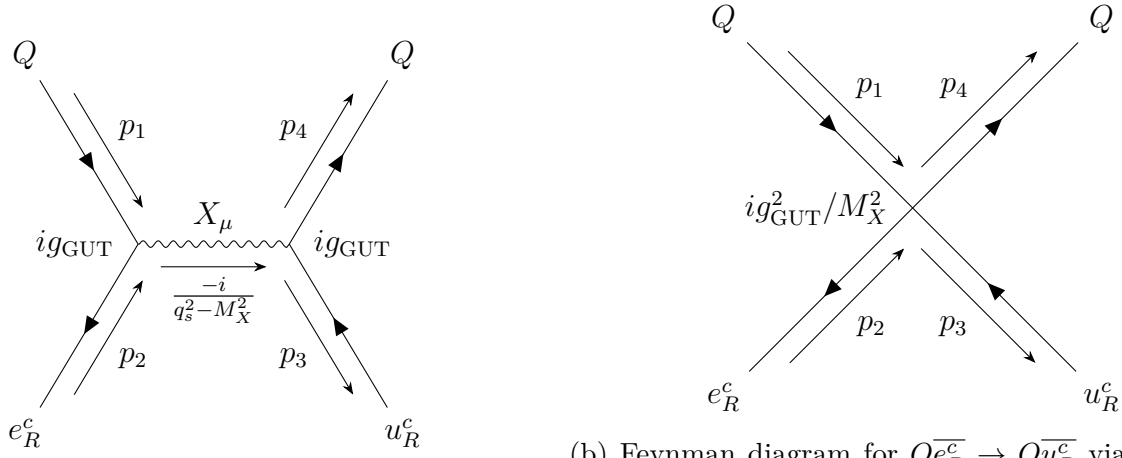
$$\alpha_3^{-1}(\mu) = 8.5 + \frac{7}{2\pi} \ln\left(\frac{\mu}{M_Z}\right) \quad (489)$$

$$= \frac{7}{2\pi} \ln(\mu) - \frac{7}{2\pi} \ln(M_Z) + 8.5. \quad (490)$$

From the sketch in Fig. 5, we can see that the three couplings do not unify at a single scale in the Standard Model. The scale at which α_1 and α_2 meet is approximately 10^{13} GeV.

(b)

The schematic couplings $g_{\text{GUT}} X_\mu Q^\dagger \bar{\sigma}^\mu e_R^c$ and $g_{\text{GUT}} X_\mu u_R^c \bar{\sigma}^\mu Q$ are consistent with the Standard Model quantum numbers of the heavy gauge boson X_μ . The first coupling describes the interaction between the heavy gauge boson X_μ , the left-handed quark doublet Q , and the right-handed charged lepton e_R^c . The second coupling describes the interaction between the heavy gauge boson X_μ , the right-handed up-type quark u_R^c , and the left-handed quark doublet Q . Integrating out the heavy X boson of mass



(a) Feynman diagram for $Q\bar{e}_R^c \rightarrow Q\bar{u}_R^c$ via s-channel interaction. This diagram is obtained by integrating out the heavy X boson in the s-channel diagram in Fig. 6a.
(b) Feynman diagram for $Q\bar{e}_R^c \rightarrow Q\bar{u}_R^c$ via a contact interaction.

Figure 6: Feynman diagrams for $Q\bar{e}_R^c \rightarrow Q\bar{u}_R^c$ scattering via gauge interactions.

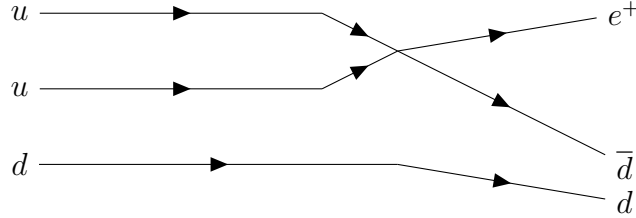


Figure 7: Feynman diagram for proton decay $p \rightarrow e^+\pi^0$ mediated by the dimension-6 operator $(QQ)(e_R^c u_R^c)$. The initial state contains two up quarks and one down quark, which can be found inside the proton. The final state contains a positron e^+ and $d\bar{d}$ quark pair, which can hadronize into a π^0 .

M_X generates a low-energy dimension-6 operator of the form

$$\mathcal{L}_{\text{eff}} \supset \frac{g_{\text{GUT}}^2}{M_X^2} (QQ)(e_R^c u_R^c), \quad (491)$$

up to contractions. Figure 7 shows a Feynman diagram for proton decay $p \rightarrow e^+\pi^0$ via the dimension-6 operator $(QQ)(e_R^c u_R^c)$. This process is baryon-number violating because the initial state contains three quarks (baryon number $B = 1$) while the final state contains no quarks (baryon number $B = 0$). Hence, this operator can mediate proton decay.

(c)

First, consider the dimension-6 operator $(QQ)(e_R^c u_R^c)$, which has a coefficient of g_{GUT}^2/M_X^2 . The decay amplitude $\mathcal{M}$ for proton decay is proportional to this coefficient, so we have

$$\mathcal{M} \sim \frac{g_{\text{GUT}}^2}{M_X^2}. \quad (492)$$

The decay rate Γ_p is proportional to the square of the amplitude, so we have

$$\Gamma_p \propto |\mathcal{M}|^2 \sim \frac{g_{\text{GUT}}^4}{M_X^4} \sim \alpha_{\text{GUT}}^2 \frac{1}{M_X^4}. \quad (493)$$

Since the decay rate is dimension of mass, we need to multiply by a factor of M_p^5 to get the correct dimensions, where M_p is the proton mass. Hence, we have

$$\Gamma_p \sim \alpha_{\text{GUT}}^2 \frac{M_p^5}{M_X^4}. \quad (494)$$

The proton lifetime τ_p is the inverse of the decay rate, so we have

$$\tau_p \sim \frac{1}{\Gamma_p} \sim \frac{M_X^4}{\alpha_{\text{GUT}}^2 M_p^5}. \quad (495)$$

Now, we can estimate the proton lifetime for $M_X \approx 10^{15}$ GeV, taking $M_p = 1$ GeV and $\alpha_{\text{GUT}} \approx 1/25$. We have

$$\tau_p \sim \frac{(10^{15} \text{ GeV})^4}{(1/25)^2 (1 \text{ GeV})^5} \quad (496)$$

$$= 25^2 \times 10^{60} \text{ GeV}^{-1} \quad (497)$$

$$\approx 6.25 \times 10^{62} \text{ GeV}^{-1} \quad (498)$$

$$\approx 6.25 \times 10^{62} \times 6.58 \times 10^{-25} \text{ s} \quad (499)$$

$$\approx 4.11 \times 10^{38} \text{ s} \quad (500)$$

$$\approx 1.30 \times 10^{31} \text{ years}. \quad (501)$$

This estimated proton lifetime is much shorter than the current lower limit $\tau(p \rightarrow e^+ \pi^0) \geq 10^{34}$ years, which means that if M_X were around 10^{15} GeV, we would have already observed proton decay.

Finally, we can use the scaling result to explain why the lower bound on M_X depends only weakly on the experimental lower bound on the proton lifetime. Since $\tau_p \sim M_X^4$, we have

$$M_X \sim \tau_p^{1/4}. \quad (502)$$

This means that if the experimental lower bound on the proton lifetime increases by a factor of 10^n , the lower bound on M_X only increases by a factor of $10^{n/4}$. For example, if the lower bound on the proton lifetime increases from 10^{34} years to 10^{35} years (an increase by a factor of 10), the lower bound on M_X only increases by a factor of $10^{1/4} \approx 1.78$. Specifically, to satisfy the current lower bound on the proton

lifetime, we need

$$M_X > 10^{15} \text{ GeV} \times 1.78^{(34-31)} \quad (503)$$

$$\approx 10^{15} \text{ GeV} \times 1.78^3 \quad (504)$$

$$\approx 10^{15} \text{ GeV} \times 5.63 \quad (505)$$

$$\approx 5.63 \times 10^{15} \text{ GeV}. \quad (506)$$

□